

A Project Report on

ECG SIGNAL EXTRACTION AND DENOISING USING EMPIRICAL MODE DECOMPOSITION

Submitted in partial fulfillment of the requirements for the award of the Degree of

Bachelor of Technology
in
Electronics and Communication Engineering

By

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(An Autonomous Institution)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada, Accredited by NBA (Tier I) and NAAC with 'A++' Grade)

Aditya Nagar, ADB Road, Surampalem

2020 – 2024

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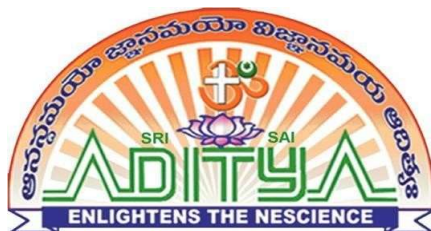
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2020 – 2024

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



CERTIFICATE

This is to certify that the thesis entitled “**ECG SIGNAL EXTRACTION AND DENOISING USING EMPIRICAL MODE DECOMPOSITION**” is being submitted by

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in partial fulfillment of the requirements for the award of degree of **B.Tech** in Electronics and Communication Engineering from **Jawaharlal Nehru Technological University, Kakinada** is a record of bonafide work carried out by them at Aditya Engineering College(A).

The results embodied in this Project report have not been submitted to any other University or Institute for the award of any degree or diploma.

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(Dr. P Sunitha)

HEAD OF THE DEPARTMENT
(Dr. N Radha)

EXTERNAL EXAMINER

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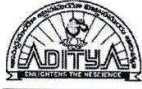
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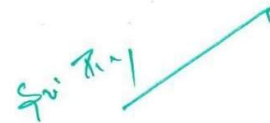
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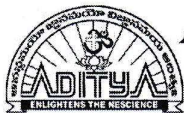
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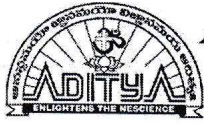
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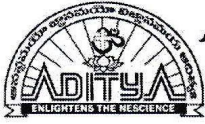
PSO1:

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PSO2:

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CO4: Analyze the obtained solution within the context of an engineering framework that addresses societal and environmental concerns while adhering to professional ethics.
CO5: Prepare a technical report with effective written communication skills.
CO6: Interpret the results of project work with oral communication skills.

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Project - PO Mapping

A.Y: 2023-24 (AR 20)

Sem.: VIII

Section & Batch:A1

Project Title: **ECG SIGNAL EXTRACTION AND DENOISING USING EMPIRICAL
MODE DECOMPOSITION**

Abstract

An Empirical Mode Decomposition based approach for ECG signal extraction is described with the goal of minimizing noise in the resulting ECG signal. The methodology involves decomposing the complex ECG signal into simpler components i.e., Intrinsic Mode Functions (IMFs). After that, the IMFs will be used to determine whether or not the generated IMF is associated with the input signal by looking up the stopping criterion. The specific IMF will be referred to as a residue if the criterion is satisfied otherwise, a new IMF will be computed until the input and IMF correlation meets the stopping criteria. The result will then be validated and compared with existing Wavelet and SVD methods using the output SNR and MSE and Accuracy. Our aim is that our approach will yield the better SNR and MSE rate and Accuracy compared to the existing method. By employing EMD, researchers and healthcare professionals can improve the detection of abnormalities and facilitate better patient care.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Overall Mapping	3	2	3	2	3	2	1	2	3	2	2	1

	PSO1	PSO2
Overall Mapping	3	2

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NOMENCLATURE

ECG	Electrocardiogram
EKG	Elektrokardiogramm (German Spelling)
DWT	Discrete Wavelet Transform
RMSE	Root Mean Square Error
MMSE	Minimum Mean Square Error
EMD	Empirical Mode Decomposition
SNR	Signal-to-noise ratio
MATLAB	Matrix Laboratory
GUI	Graphical User Interface
FFT	Fast Fourier Transform
SVD	Singular Value Decomposition
MSE	Mean Square Error
CTM	Central Tendency Machine
PLI	Power Line Interference
WPT	Wavelet Packet Transform
PNN	Probabilistic Neural Network
MIV	Mean Impact Value
ICA	Independent Component Analysis
IMF	Intrinsic Mode Function
CPET	Cardio Pulmonary Exercise Test
CTA	Computed Tomography Angiography
MRA	Magnetic Resonance Angiography

Abstract

An Empirical Mode Decomposition based approach for ECG signal extraction is described with the goal of minimizing noise in the resulting ECG signal. The methodology involves decomposing the complex ECG signal into simpler components i.e., Intrinsic Mode Functions (IMFs). After that, the IMFs will be used to determine whether or not the generated IMF is associated with the input signal by looking up the stopping criterion. The specific IMF will be referred to as a residue if the criterion is satisfied otherwise, a new IMF will be computed until the input and IMF correlation meets the stopping criteria. The result will then be validated and compared with existing Wavelet and SVD methods using the output SNR and MSE and Accuracy. Our aim is that our approach will yield the better SNR and MSE rate and Accuracy compared to the existing method. By employing EMD, researchers and healthcare professionals can improve the detection of abnormalities and facilitate better patient care.

CHAPTER-1

INTRODUCTION

1.1 SIGNAL :

Anything that carries information can be called as signal. It can also be defined as a physical quantity that varies with time, temperature, pressure or with any independent variables such as speech signal or video signal. The process of operation in which the characteristics of a signal Amplitude, Shape, Phase, frequency, etc. undergoes a change is known as signal processing.

Any unwanted signal interfering with the main signal is termed as noise. So, noise is also a signal but unwanted. According to their representation and processing, signals can be classified into various categories.

- Continuous Time Signals
- Discrete Time Signals

1.1.1 Continuous Time Signals:

Continuous-time signals are defined along a continuum of time and are thus, represented by a continuous independent variable. Continuous-time signals are often referred to as analog signals. This type of signal shows continuity both in amplitude and time. These will have values at each instant of time. Sine and cosine functions are the best example of Continuous time signal.

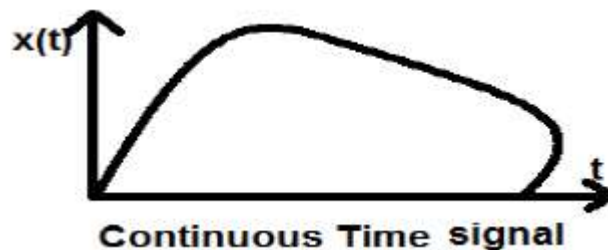


Fig 1.1 : Continuous Time Signal

The signal shown above is an example of continuous time signal because we can get value of signal at each instant of time.

1.1.2 Discrete Time Signals

The signals, which are defined at discrete times are known as discrete signals. Therefore, every independent variable has distinct value. Thus, they are represented as sequence of numbers. Although speech and video signals have the privilege to be represented in both continuous and discrete time format; under certain circumstances, they are identical. Amplitudes also show discrete characteristics. Perfect example of this is a digital signal; whose amplitude and time both are discrete.

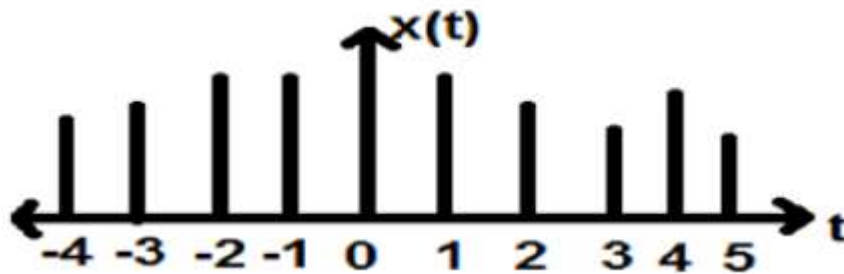


Fig 1.2 : Discrete Time Signal

The figure above depicts a discrete signal's discrete amplitude characteristic over a period of time. Mathematically, these types of signals can be formularized as;

$$x = \{x[n]\}, -\infty < n < \infty \quad 1.1$$

Where, n is an integer.

It is a sequence of numbers x , where n^{th} number in the sequence is represented as $x[n]$.

1.1.3 Digital Signal Processing

Digital Signal Processing (DSP) is a branch of engineering and applied mathematics that deals with the processing and analysis of digital signals. A digital signal is a discrete-time signal, that is represented by a sequence of numbers sampled at regular intervals. DSP involves various algorithms, techniques, and methodologies to process these digital signals to retrieve essential information or improve specific features.

Digital Signal processing, officially, is quite complex. It can standardize or solve digital signals, but it can also carry out a variety of other functions, including filtering, compression and modulation. DSP algorithms are capable of distinguishing between

orderly signals and noise, although they may not always achieve perfect results.

Communication system involves noise levels, irrespective of whether the signals carry both analog and digital in spite of the type of information is transmitted. Noise exhibits a continuous challenge in digital signal processing to improve the Signal-to-noise(S/N) ratio in DSP. To improve the improve an efficiency of the S/N ratio that involves the improvement of transmitted signal power and increases receiver sensitivity.

Digital signal processor has some advantages.

- It includes digital signal which has a less probability of getting mixed with unwanted signals so that overall noise will be less.
- It allows usage of numerous error detection and correction characteristics that exists to digital signals such as a detection and correction tool and utilizes a parity generation and correction.
- It is used to store digital data in a simple way. It is required to select from a different variety of digital memories. Digital signals are involved in simple encryption.

Digital signal processor has some disadvantages.

- DSP system has some complexities that leads to increase sue to the use of additional components.
- Digital signal processor utilizes various transistors which needs huge power compared to the analog signal processors.
- Digital signal processors are very expensive. Digital communications use wide range of bandwidth to send the data compared to analog method.
- It samples analog signals and quantizing them into digital signal can give errors, gives attenuation with respect to information in the signal.

DSP is a fundamental technology that has revolutionized the way, manipulate, and analyze digital signals across various domains. Using some computational algorithms and techniques, DSP gives flexibility that matches to basic analog signal processing methods.

1.2 Electrocardiogram (ECG):

One of the first biological signals used in electronic and clinical medicine was the ECG signal. Unlike other biological signs, the ECG signal is simpler to recognize. The ECG signal is typically weak and vulnerable to environmental perturbations during the ECG signal acquisition, digital-to-analog conversion, and storage processes. The extraction of the ECG signal is therefore essential in signal processing. Reducing noise in the ECG signal is essential for signal recovery. The filter approach, empirical mode decomposition method, neural network method, and wavelet transform method are a few examples of common denoising techniques for ECG signals.

When electrocardiography is performed, an electrocardiogram (ECG or EKG [a]), a recording of the electrical activity of the heart during repeated cardiac cycles, is produced. It is an electrogram of the heart, which plots the voltage of the heart's electrical activity against time using electrodes placed on the skin. These electrodes detect the minute electrical changes that result from the cardiac muscle's periodic depolarization and repolarization throughout each cardiac cycle (heartbeat).

The normal ECG pattern can be altered by a variety of cardiac abnormalities, including irregular heartbeat (like atrial fibrillation and ventricular tachycardia), insufficient coronary artery blood flow (like myocardial ischemia and myocardial infarction), and electrolyte abnormalities (like hypokalemia and hyperkalemia). Traditionally, "ECG" has referred to a 12-lead ECG that was recorded while lying down, as explained below. Although some kinds of smartwatches are also able to record an ECG, other devices, like a Holter monitor, can record the electrical activity of the heart. ECG signals can be captured using several methods and circumstances.

Ten electrodes are positioned on the patient's limbs and the surface of the chest during a standard 12-lead ECG. Then, over a period of time (often 10 seconds), the overall magnitude of the heart's electrical potential is measured from twelve various angles (referred to as "leads"). Thus, at each instant of the cardiac cycle, the total magnitude and direction of the heart's electrical depolarization are recorded. There are three main components to an ECG: the P wave, which represents depolarization of the atria; the QRS complex, which represents depolarization of the ventricles; and the T wave, which represents repolarization of the ventricles.

A healthy heart depolarizes in an orderly manner throughout the atrium, sinoatrial node, atrioventricular node, and bundle of his, Purkinje fibres, and down and to the left throughout the ventricles during each heartbeat. This depolarization is initiated by pacemaker cells in the atrium. The distinctive ECG tracing results from this systematic pattern of depolarization. An ECG provides a highly detailed picture of the anatomy of the heart and the operation of its electrical conduction system to the experienced clinician. An ECG can be used to determine the frequency and rhythm of heartbeats, the size and location of the heart chambers, the presence of any cardiac muscle cell damage, or conduction abnormalities, among other things system, the effects of heart drugs, and the function of implanted pacemakers.

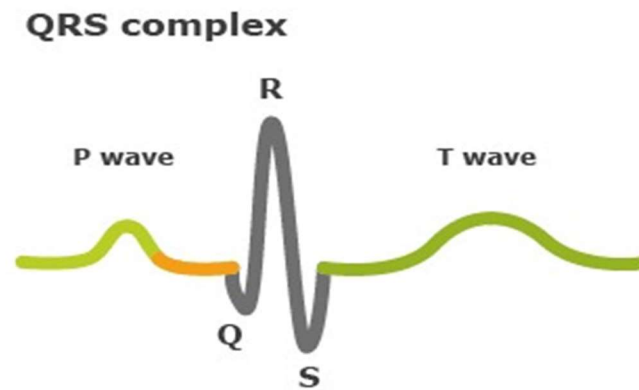


Fig 1.3: An ECG Wave showing all its Peaks

1.2.1 Characteristics of an ECG Signal

Three deflections in the ECG waveform make up the QRS complex. The most noticeable component of the human ECG is the QRS complex, which depicts the depolarization of the right and left ventricles. Load and plot an ECG waveform with two or more cardiologist-annotated R peaks of the QRS complex. The QRS complex is a key component of an electrocardiogram (ECG) waveform, representing the electrical activity associated with ventricular depolarization. Here are some important characteristics of the QRS complex. The duration of the QRS complex is typically measured from the beginning of the Q wave to the end of the S wave. Normal duration is usually less than 0.12 seconds.

Table 1.1: Amplitude and Duration of ECG Signal

Amplitude(v)	Duration(Seconds)
P wave - 0.25mV	PR interval - 0.12s to 0.20s
R wave - 1.60mV	QT interval - 0.35s to 0.44s
Q wave - 25% of R wave	ST interval - 0.05s to 0.15s
T wave - 0.1 to 0.5mV	QRS interval – 0.09s

Three of the visual deflections present on a typical electrocardiogram (ECG or EKG) are combined to form the QRS complex. It is typically the most prominent and central area of the trace. It is the same as the heart's right and left ventricles depolarizing and contracting their big ventricular muscles. The QRS complex typically lasts 80 to 100 ms in adults; it may be shorter in children. The Q, R, and S waves all follow one another quickly, represent the same occurrence, and do not all show in all leads. For these reasons, they are frequently grouped together. Any downward deflection that comes right after the P wave is known as a Q wave. Following an upward deflection known as a R wave, the S wave is any subsequent downward deflection.

The S wave is followed by the T wave, and in some circumstances, a subsequent U wave. Start from the end of the PR interval (or the beginning of the Q wave) and go to the end of the S wave to calculate the QRS interval. This gap typically lasts 0.08 to 0.10 seconds. A wide QRS complex is one whose duration is greater. It comprises multiple waves and complexes, including the P wave, representing atrial depolarization, the QRS complex, signifying ventricular depolarization and contraction, and the T wave, indicating ventricular repolarization. Analyzing an ECG involves examining the duration, amplitude, and morphology of these waves, along with factors like heart rate, rhythm regularity, and axis deviation.

1.3 Applications of Digital Signal Processing :

- Audio Processing
- Image Processing
- Speech Processing
- Wireless Communication
- Radar and Sonar Processing
- Biomedical Signal Processing
- Control Systems
- Sensor signal processing
- Audio and Video Compression
- Geophysical Processing

1.4 Software Requirement :

MATLAB, Version: R2022a,64-bit

1.5 Motivation :

ECG signals contain vital information about cardiac health and function, making them indispensable for diagnosing various cardiac conditions. Extracting features from ECG signals, provides valuable insights into cardiac abnormalities and helps in clinical decision-making. Accurate detection and analysis of ECG features are crucial for diagnosing arrhythmias, myocardial infarction, and other cardiac disorders. Feature extracting using EMD can improve the accuracy of ECG analysis by identifying key signal components and reducing the influence of noise and artifacts.

ECG signals are often corrupted by various types of noise, including baseline wander, muscle artifacts, electrode motion artifacts, and electromagnetic interference. Denoising ECG signals using EMD can effectively separate noise components from the underlying signal, thereby enhancing signal quantity and improving diagnostic accuracy. Researchers have presented several methods for denoising and extracting features from the ECG signal.

The study of the electrical activity of the heart traces back to the late 19th century. Initial ECG recordings were made using cumbersome equipment and paper-based chart

recorders, making signal analysis and interpretation challenging. Manual methods for ECG signal extraction and denoising relied on human expertise and were prone to subjective interpretation and errors. The advent of DSP techniques in the late 20th century revolutionized ECG signal analysis and interpretation. Computerized algorithms were developed for automated ECG signal processing, including feature extraction, denoising and arrhythmia detection.

1.6 Need for Extracting Features and Denoising ECG Signal :

ECG signals provide crucial information about the electrical activity of the heart. Extracting features such as R peaks, QRS complexes, ST segments, and T waves helps clinicians diagnose various cardiac conditions, including arrhythmias, myocardial infarction, heart rate variability. Monitoring changes in these features over time can provide insights into the progression of cardiac disorders and the effectiveness of treatment.

Extracting ECG features can inform treatment planning and management decisions. Features extraction allows for the customization of treatment strategies based on individual patient characteristics. By analyzing ECG features along with other clinical data, healthcare providers can tailor treatment plans to the specific needs and risk profiles of each patient.

ECG signals are often corrupted by various types of noise, including baseline wander, power line interference, muscle artifacts, electrode motion artifacts and electromagnetic interference. Denoising ECG signals removes these unwanted noise components, allowing clinicians to accurately identify and analyze key waveform features. By reducing noise interference, denoising enhances the accuracy of diagnostic algorithms and improves the reliability of clinical diagnoses.

Clean and reliable ECG signals facilitate more precise measurements of waveform characteristics such as R peaks, QRS complexes, ST segments and T waves. Enhanced signal quality ensures that important waveform features are accurately represented, enabling clinicians to make informed decisions based on reliable data.

Noisy ECG signals can lead to false alarms in monitoring systems, triggering unnecessary interventions or alerts. Denoising ECG signals reduces the occurrence of false alarms by eliminating noise artifacts and ensuring that alarms are triggered only in response to clinically relevant events. Many signal processing algorithms used for ECG analysis,

such as feature extraction and arrhythmia detection are sensitive to noise. Denoising ECG signals improves the performance of these algorithms by providing cleaner input data, leading to more accurate and reliable results.

1.7 Problem Statement :

Electrocardiogram (ECG) signals are fundamental in diagnosing various cardiac conditions, including arrhythmias, myocardial infarction, and heart failure. However, the accurate extraction of meaningful features from ECG signals is often hindered by noise interference. Sources of noise include baseline wander, powerline interference, muscle artifacts, electrode motion, and electromagnetic interference. These noise components can obscure the underlying cardiac information, leading to misinterpretation and inaccurate diagnosis.

The challenge lies in developing robust signal processing techniques capable of separating the desired ECG components from noise effectively. Existing methods, such as filtering and Fourier-based approaches, often exhibit limitations in handling non-stationary signals and may inadvertently remove important signal components along with noise. Thus, there is a pressing need for advanced techniques that can accurately extract ECG features while mitigating the effects of noise interference.

Extracting features from electrocardiogram (ECG) signals using Singular Value Decomposition (SVD) and wavelet transforms faces several challenges. Firstly, these methods may struggle with noise interference, hindering accurate feature extraction. Additionally, the complex characteristics of ECG signals, such as non-stationarity and dynamic frequency components, pose difficulties for traditional techniques.

Moreover, wavelet transforms may lack precise frequency localization, impacting the extraction of vital signal details. Furthermore, the computational complexity of these methods limits their applicability in real-time or resource-constrained environments. Finally, there's a need for adaptive techniques capable of adjusting to the diverse characteristics of ECG signals and noise sources encountered in practical settings. Overcoming these challenges is crucial for improving the accuracy and efficiency of ECG analysis in clinical practice and biomedical research.

1.8 Objectives and Goals:

The primary objective of this project is to address the challenges encountered in electrocardiogram (ECG) feature extraction using Singular Value Decomposition (SVD) and wavelet techniques, and to establish the superiority of the Empirical Mode Decomposition (EMD) method over traditional approaches. Specifically, we aim to overcome limitations such as noise interference and lack of adaptability inherent in SVD and wavelet-based methods.

Through comprehensive experimentation and analysis, we will evaluate the performance of EMD against traditional methods based on signal-to-noise ratio (SNR), mean squared error (MSE), and feature extraction accuracy metrics. By demonstrating the efficacy of EMD in mitigating noise and preserving essential signal characteristics, we seek to provide compelling evidence for its superiority as a feature extraction technique for ECG signals. This project will contribute to advancing the understanding of ECG signal processing techniques and lay the groundwork for future research in cardiac diagnostics and healthcare applications.

1.9 Organization of Thesis

In **chapter-1** it explains Introduction to ECG and DSP and its applications.

In **chapter-2** it explains literature survey, different methods in extracting features of ECG and denoising like SVD, wavelet etc., and also includes existing methods and its drawbacks.

In **chapter-3** it explains about the ECG signal and its peaks, machines.

In **chapter-4** the proposed algorithm, the major aim of the project is explained.

In **chapter-5** it introduces implementation of proposed work.

In **Chapter-6** it discusses the performance characteristics of ECG based on EMD and compare its results with existing method.

In **Chapter-7** it concludes the thesis and future scope of project further research.

CHAPTER-2

LITERATURE SURVEY

2.1 Introduction :

In order to understand the current landscape of ECG signal processing techniques and to identify the potential advantages of employing Empirical Mode Decomposition (EMD) for signal extraction and denoising a comprehensive literature survey was conducted. This survey aimed to explore existing methodologies, algorithms, and approaches utilized in the field of ECG signal processing, with a specific focus on noise reduction techniques and feature extraction methods. By examining the strengths and limitations of various approaches, including wavelet transform, singular value decomposition(SVD), and other advanced signal processing techniques, the literature survey provided valuable insights into the challenges faced by researchers and healthcare professionals in analyzing ECG data accurately. Furthermore, the survey aimed to identify gaps in the existing literature and opportunities for innovation, particularly in the context of enhancing the diagnostic accuracy and clinical utility of ECG analysis. Through a systematic review of relevant research articles, conference papers, and academic publications, this literature survey laid the foundation for the development of a novel EMD-based approach for ECG signal Processing, with the goal of minimizing noise and improving the detection of cardiac abnormalities.

2.2 Research papers :

S. K . Sahoo, 2015 et al. explained automating the detection of cardiac arrhythmias from ECG signals using wavelet transform-based techniques. Cardiac arrhythmia, characterized by abnormal electrical activity in the heart, can be analyzed by extracting morphological features from ECG signals. The study aims to detect arrhythmias by identifying the QRS complex and R-peak through wavelet transform analysis. Data from the MIT-BIH arrhythmia database was utilized for evaluation. The paper describes the implementation of different peak and QRS complex detection algorithms based on Discrete Wavelet Transform (DWT), comparing results for normal and abnormal ECG signals. Wavelet analysis aids in identifying abnormalities reflected in the detailed coefficients of the ECG signal.

The performance of DWT-based detectors was thoroughly examined, demonstrating superior morphological values for normal ECG signals in terms of amplitude and QRS duration [1].

A.Ekhlasi, 2022 et al. proposed the application of Empirical Mode Decomposition (EMD) for analyzing Electro-Cardiograph (ECG) signals and detecting Atrial Fibrillation(AF) cases. In this study, IMFs are obtained using the Hilbert transform to derive analytic functions. The Central Tendency Measure (CTM) is then employed to determine a radius within which 95% of the analytic function points lie, serving as a feature for categorizing normal and AF ECG signals. The paper's main focus lies in differentiating between normal and AF ECG signals using the proposed method. By computing the area feature corresponding to the radius of the circle in the complex plane for each signal, significant differences between normal and AF ECG signals are observed. Specifically, the results indicate that normal ECG signals exhibit larger area criterion values compared to AF signals [2].

Al-Zuhairi, 2022 et al. introduces a novel algorithm designed to remove muscle artifact, electrode motion noise, and additive white Gaussian noise commonly present in ECG signals. While singular value decomposition (SVD) has been used for denoising, residual noise remains due to differing strategies for determining signal and noise eigen components. The proposed algorithm employs SVD in double stages: initially, a regular SVD divides data into overlapped groups, each arranged in a Hankel matrix to expand processing space. Subsequently, SVD is applied again on each Hankel matrix to further reduce noise. Simulation results using input ECG signals at 5 dB and 10 dB signal-to-noise ratios (SNRs) demonstrate significant improvement in SNR, mean square error (MSE), and percent root mean square difference (PRD), with the proposed algorithm achieving superior performance compared to existing denoising techniques [3].

H.-Y. Lin, 2014 et al. proposed preprocessing approach using discrete wavelet transform (DWT) to remove baseline wander and wavelet packet transform (WPT) to eliminate power line interference (PLI) from ECG signals. Following noise removal, DWT is utilized to analyze noise-free ECGs, generating approximation and detail coefficients. Using second detail coefficients obtained from reconstructed signals of the MIT-BIH

Database, Haar wavelet-based analysis achieves high sensitivity, positive, predictivity, predictivity, and low detection error rates for R-peak detection. The method employing discrete wavelet transform significantly enhances the detection of onset and offset locations and durations of P, QRS, and T waves in denoised ECG signals. The use of DWT with Haar mother wavelet for analyze in reconstructed signals improves the detection accuracy of R-peaks. Additionally, preprocessing with DWT and WPT effectively removes baseline wander and power line interference, leading to slight improvements in the detection of wave onset and offset [4].

Li, N, 2022 et al. introduces a novel ECG signal identification method integrating the wavelet transform algorithm with the probabilistic Neural Network (PNN) optimized by the Whale Optimization Algorithm. Initially, the Q, R, and S waves are detected using wavelet transform, while the P and T waves are identified through local windowed wavelet transform, constructing characteristic values based on detected time points to reduce ECG data dimensionality. Subsequently, the PNN, combined with the mean impact value (MIV) algorithm, filters out low-impact characteristic values, simplifying model input and complexity. The study signifies the increasing adoption of ECG signal biometric technology over traditional recognition methods. The proposed method, leveraging wavelet transform and WOA-PNN algorithms, demonstrates robust performance in ECG signal identification [5].

Rahangdale, 2018 et al. focuses on the computation of feature extraction from ECG signals and evaluates its performance. The Empirical Mode Decomposition (EMD) and Wavelet Transform are identified as the most suitable techniques for analyzing ECG signals, with Wavelet Transform being particularly adept for non-stationary signal analysis. This paper addresses the crucial task of extracting features from ECG signals, which is essential for diagnosing coronary diseases and understanding cardiac health. It highlights the importance of accurate peak detection, particularly in the QRS complex, and emphasizes the significance of employing suitable techniques like EMD and Wavelet Transform for signal analysis. The study utilizes MATLAB Programming for analysis and evaluates its performance using the MIT-BIH ECG database [6].

Peri, E, 2021 et al. introduces a novel algorithm based on SVD for removing cardiac

contamination from trunk electromyography(EMG) signals. The algorithm's performance is evaluated across various signal-to-noise ratios (SNRs) and compared with existing methods such as gating, high-pass filtering, template subtraction (TS), and independent component analysis (ICA). A synthetic dataset combining electrocardiography (ECG) and EMG is generated to establish a reference for validation. The paper presents a novel method for removing ECG interference from EMG recordings, demonstrating superior performance in the time domain and comparable results in the frequency domain compared to existing approaches. Application of the algorithm on real data shows promise in estimating respiratory patterns from diaphragm EMG signals, with the added advantage of not requiring an additional reference ECG [7].

Bandarabadi, 2010 et al. proposed a method for optimizing electrocardiogram (ECG) signals by filtering singular values. The proposed method demonstrates a significant capability in enhancing ECG signals, particularly in the presence of additive white noise. Signals with varying levels of noise are evaluated based on signal-to-noise ratio (SNR), and their time-frequency spectra are compared. The study presents a new method for noise reduction in ECG signals by optimizing singular values. Through quantitative and qualitative evaluations, the efficacy of the method is demonstrated [8].

M. A. Ouali, 2013 et al. proposed an efficient filtering technique based on Singular Value Decomposition (SVD) for denoising electrocardiogram (ECG) signals. SVD is employed as a high-resolution spectrum estimation tool to decompose the ECG data matrix into orthogonal subspaces. These subspaces represent the signal and noise components present in the ECG data. By projecting the data onto the desired subspace, the noise and unwanted signal components are effectively eliminated. The study utilizes Singular Value Decomposition for denoising ECG signals using both simulated and recorded data. By leveraging SVD, the noise and genuine ECG signal are separated into dominant and subdominant subspaces, respectively. Simulation results demonstrate the effectiveness of this method, particularly when a suitable projection operator is chosen [9].

Jianjun He, 2015 et al. addressed the challenge of accurately and efficiently extracting features from individual electrocardiogram (ECG) signals for authentication purposes. Traditional time-frequency analysis methods like Fourier and wavelet transforms face

limitations when dealing with non-stationary signals, leading to false signal production and aliasing. To overcome these limitations, the paper proposes the use of Empirical Mode Decomposition (EMD), which is effective for extracting non-stationary and nonlinear signals. EMD decomposes the ECG signal into a series of basic mode components and a remainder term. Unlike Fourier analysis, EMD relies solely on the local time scale of the signal without any predefined basis functions. The paper discusses an improved EMD algorithm and an enhanced mirror extension method, which mitigate the effects of end effects, thereby improving the accuracy of EMD [10].

Hussein, 2022 et al. introduces an adaptive Electrocardiogram (ECG) denoising technique based on Empirical Mode Decomposition (EMD). The primary objective is to enhance traditional EMD-based ECG denoising methods by adaptively calculating lower Intrinsic Mode Function (IMF) mean values during signal reconstruction. Unlike conventional approaches that often discard initial IMFs using a window-based method to mitigate high-frequency noise, the proposed method aims to preserve high-frequency detail while accurately removing noise, particularly in the QRS complex region. The proposed technique is evaluated using ECG data from the MIT-BIH and Brno University of Technology ECG Quality Database (BUT QDB). A comprehensive simulation study is conducted by introducing various noise types, including Gaussian white noise and electromyography (EMG) noise, at different signal-to-noise ratio (SNR) levels. The performance of the proposed method is assessed using three key evaluation metrics: Signal-to-Noise Ratio (SNR), Percent Root Mean Square Deviation (PRD), and Mean Square Error (MSE) [11].

L. Zebin, 2020 et al. addresses the challenge of extracting effective Electrocardiogram (ECG) signals from strong background interference and noise, which is crucial for accurate diagnosis of various cardiac conditions such as arrhythmia and myocardial infarction. After evaluating several denoising methods, the paper proposes a novel approach combining Empirical Mode Decomposition (EMD) with wavelet soft thresholding. The proposed method involves decomposing the ECG signal into multi-layer Intrinsic Mode Function (IMF) modal components using EMD. Subsequently, the high-frequency IMF modal component, containing noise, is processed using wavelet thresholding to extract the useful

ECG signal component. Finally, the denoised ECG signal is reconstructed using the extracted useful signals from high-frequency IMFs and the modal components of low-frequency IMFs. The results demonstrate that the proposed EMD combined with wavelet soft threshold method outperforms other denoising techniques such as EMD, wavelet transform, and band-pass filtering in terms of Signal-to-Noise Ratio (SNR), Root Mean Square Error (RMSE), and coefficient correlation [12].

Y. Lu, 2009 et al. proposes a novel scheme for denoising Electrocardiogram (ECG) signals by integrating an ECG dynamic model with Empirical Mode Decomposition (EMD). The method aims to effectively remove noise while preserving important morphological features, particularly the QRS complex. The proposed method begins by pre-filtering the noisy ECG signal using a dynamic model to fit it in the Minimum Mean Square Error (MMSE) sense. This step focuses on preserving crucial morphological features, especially the QRS complex. Subsequently, the model is subtracted from the noisy ECG, leaving a residual signal. The residual signal is then decomposed using EMD, and the noise components are discarded from the decomposition results.

Finally, the denoised ECG signal is obtained by combining the model and the denoised residue. Experimental evaluations conducted on both real and synthetic ECG data demonstrate the superiority of the proposed method for ECG denoising. The paper concludes that while EMD is an effective tool for analyzing nonstationary and nonlinear data, direct EMD denoising is not ideal for ECG signals due to the high and sharp R waves, which can be easily deformed. By incorporating an ECG model into the denoising process, the proposed method effectively preserves the characteristic waves of the ECG signal while reducing noise. Overall, the method offers simplicity, flexibility, and efficacy for ECG denoising applications[13].

F. Luo, 2023 et al. proposed method begins with singular value decomposition (SVD) on the received ECG signal to obtain signal components corresponding to each singular value. The component corresponding to the maximum singular value is selected, and its cross-correlation coefficients with other components are calculated. The cumulative contribution rate of singular values is then used to determine the number of singular values for ECG signal reconstruction. Next, wavelet threshold denoising is applied to the selected

signal components to suppress noise effectively. Finally, the denoised ECG signal is reconstructed using the processed signal components. Experimental results demonstrate that the proposed method effectively suppresses noise and extracts the ECG signal. The analysis shows that the method outperforms the wavelet threshold method in noise reduction while preserving the ECG signal. The effective selection of singular values contributes to the successful removal of noise, resulting in a well-reserved ECG signal [14].

Zhang, 2020 et al. proposed a novel threshold denoising function and combines EMD with sample entropy to determine the order of denoised IMFs. Simulation experiments using ECG signals from the MIT-BIH database are conducted to evaluate the proposed method's performance. Various types of noise, including white Gaussian noise, electromyogram (EMG), and power line interference, are added to assess the method's robustness. Evaluation metrics such as signal-to-noise ratio improvement (SNR imp.) and mean square error (MSE) are used to compare the proposed method with existing denoising techniques. His study concludes that the proposed ECG denoising method effectively reduces noise while retaining more details of the original signal compared to existing techniques. Qualitative and quantitative analyses support the effectiveness of the proposed algorithm. [15].

2.3 Existing Method:

The non-stationary and non-linear nature of the ECG signal prevents the filter approach from solving the spectrum's issue changes with time. Additionally, model aliasing is a problem that is largely brought on by the empirical model decomposition technique. The neural network method requires a lot of experimental data, and obtaining this data might be challenging given how much different ECG signals can differ from one another. The wavelet transform's effective de-noising effect has limitations for non-uniform noise distribution, but it can handle multi-directional and multi-level dynamic analysis.

To address the shortcomings of the aforementioned approaches, a unique value selection and wavelet-based ECG signal extraction method is presented. Prior to acquiring the ECG signal component associated with each distinct value, the incoming ECG signal is first singular value decomposed. The cross-correlation between the signal component corresponding to the largest singular value and the signal component corresponding to other

singular values can then be found. The cumulative singular contribution rate is then calculated. It is combined with the cross-correlation results to determine the number of singular values needed for the reconstruction of the ECG signal. The calculated final signal component's noise is reduced via the wavelet threshold method. Finally, the ECG signal after noise reduction is obtained by reconstructing the signal component after noise reduction. It experimental results shown that the method can suppress noise and extract the ECG signal efficiently.

Suppose $X = \{x(1), x(2), \dots, x(N)\}$ is one dimensional signal, which is discrete. It is constructed as a $m \times n$ ($m \leq n$) Hankel matrix.

$$H_{m \times n} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \\ x_2 & x_3 & \dots & x_{n-1} \\ \vdots & \vdots & & \vdots \\ x_m & x_{m-1} & \dots & x_L \end{bmatrix}$$

Where H is a matrix, and m is the dimension, $m+n-1=L$. Since the ECG signal is a signal with strong periodicity, Fourier transform is performed on the ECG signal to find its period, and n is determined according to the frequency of the maximum value of the ECG signal spectrum.

$$N=1/\text{freq} * f_s \quad 2.1$$

Hankel matrix is operated by singular value

$$H=U \sum V^T \quad 2.2$$

Decomposition:

Only a single singular value λ_i is retained, and all singular values except λ_i for are set to zero, there is:

$$A_i = \begin{bmatrix} s_i & 0 \\ 0 & 0 \end{bmatrix}, s_i = \text{diag}(0, \dots, \lambda_i, 0, \dots, 0) \quad 2.3$$

$$H_i = U A_i V^T \quad 2.4$$

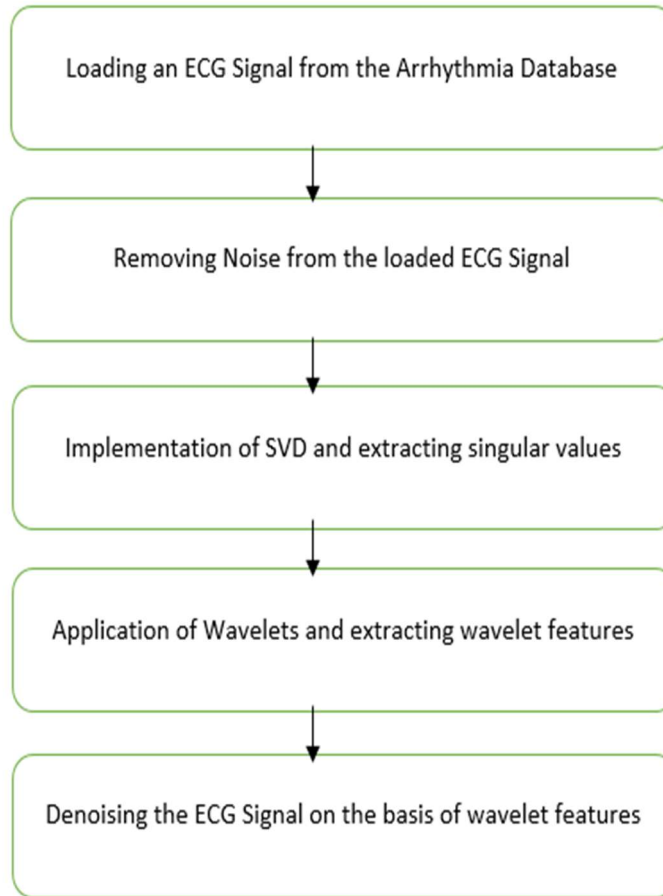


Fig 2.1: Flow of SVD And Wavelet

By using formula, a new Hankel matrix is calculated, and each singular value corresponds to an original signal component. As the reference signal, the original signal components corresponding to other singular values are cross-correlated with the reference signal, and there is:

$$R_{1i}(H_1^T H_i) = E(H_1^T H_i) = \sum_{j=-\infty}^{\infty} H_1(j) H_i(j-n) \quad 2.5$$

Where $i=2,3,\dots,N$ since the noise and the signal are independent of each other, their cross-correlation must be zero. If the component contains a signal, its cross-correlation with the reference signal must not be zero. And the correlation of the signal is closely related to time, and both positive and negative correlations indicate the degree of correlation between the component and the reference component, so the maximum value of the absolute value of the cross-correlation function is taken as the characterization.

The formula is as follows:

$$X = \max(\text{abs}(R_{1i})) \quad 2.6$$

The formulas is united to determine the number of effective singular values, and the remaining singular values are set to 0 for noise reduction.

2.3.1 WAVELET THRESHOLD DE-NOISING PRINCIPLE

Wavelet transform is developed by short-time Fourier transform. It has the characteristics of time-frequency localization and multi-resolution. There is:

$$\begin{cases} x_k^{j+1} = D(h^* x_k^j) \\ d_k^{j+1} = D(h_1^* x_k^j) \end{cases} \quad 2.7$$

Where x_k^j is the approximation signal, and d_k^j is the detail signal, and h_0 is the low pass filter, and h_1 is a band pass filter, and D is a sampling operator. Its reconstruction formula is:

$$x_n^{(j-1)} = \sum_k h_0(n-2k)x_k^{(j)} + \sum_k h_1(n-2nk)d_k^{(j)} \quad 2.8$$

Wavelet transform is an inner product operation between signal and wavelet function. It decomposes signal into different frequency band space and has multi-resolution function. It also has time-frequency localization function. Wavelet threshold is assigned to hard threshold and soft threshold. The hard threshold is to keep the point unchanged. The soft threshold is less rough than the hard threshold. The de-noising functions are as follows:

$$d_k^{j'} = \begin{cases} d_k^j & |d_k^j| > T \\ 0 & |d_k^j| \leq T \end{cases} \quad 2.9$$

$$d_k^{j'} = \begin{cases} \text{sign}(d_k^j)(|d_k^j| - T) & |d_k^j| > T \\ 0 & |d_k^j| \leq T \end{cases} \quad 2.10$$

The wavelet threshold selection rules are rigrsure, heursure, sqtwolog and minimaxi. Rigrsure is based on unbiased likelihood estimation principle. And sqtwolog uses a fixed threshold form. Heursure is the selection of the optimal predictor threshold. Minimax uses a minimax principle, which produces a minimum mean square error.

2.3.2 The ECG Extraction based on Singular Value selection and wavelet:

The procedures of the proposed method are as follows:

1. The spectrum of the ECG signal is calculated to find the frequency corresponding to the maximum spectrum, and n is determined according to Equation (2.1)
2. The singular value decomposition of the ECG signal is performed to obtain the singular values. The number of effective singular values is determined by combining formula (2.3) and formula (2.4), and the other singular values are set to 0
3. The signal corresponding to the effective singular value is processed by wavelet threshold de-noising.
4. The de-noising ECG signal is obtained by using linear superposition of the signal after wavelet threshold de-noising.

2.3.3 Disadvantages

- Noise reduction process involves two methods which makes it complex.
- Noise reduction phase need more accurate singular value decomposition for a better which is not possible.
- The wavelet decomposition sometimes loses some features of the signal.

CHAPTER-3

ECG SIGNAL

3.1 Electrocardiography:

Electrocardiography is the process of producing an electrocardiogram (ECG or EKG), a recording of the heart's electrical activity through repeated cardiac cycles. It is an electrogram of the heart which is a graph of voltage versus time of the electrical activity of the heart using electrodes placed on the skin. These electrodes detect the small electrical changes that are a consequence of cardiac muscle depolarization followed by repolarization during each cardiac cycle (heartbeat). Changes in the normal ECG pattern occur in numerous cardiac abnormalities, including:

- Cardiac rhythm disturbances (such as atrial fibrillation and ventricular tachycardia),
- Inadequate coronary artery blood flow (such as myocardial ischemia and myocardial infarction),
- and electrolyte disturbances, such as hypokalemia.

Traditionally, "ECG" usually means a 12-lead ECG taken while lying down as discussed below. However, other devices can record the electrical activity of the heart such as a Holter monitor but also some models of smartwatch are capable of recording an ECG. ECG signals can be recorded in other contexts with other devices.

In a conventional 12-lead ECG, ten electrodes are placed on the patient's limbs and on the surface of the chest. The overall magnitude of the heart's electrical potential is then measured from twelve different angles ("leads") and is recorded over a period of time (usually ten seconds). In this way, the overall magnitude and direction of the heart's electrical depolarization is captured at each moment throughout the cardiac cycle.

There are three main components to an ECG:

- The p wave, which represents depolarization of the atria.
- The QRS complex, which represents depolarization of the ventricles.
- The T wave, which represents repolarization of the ventricles.

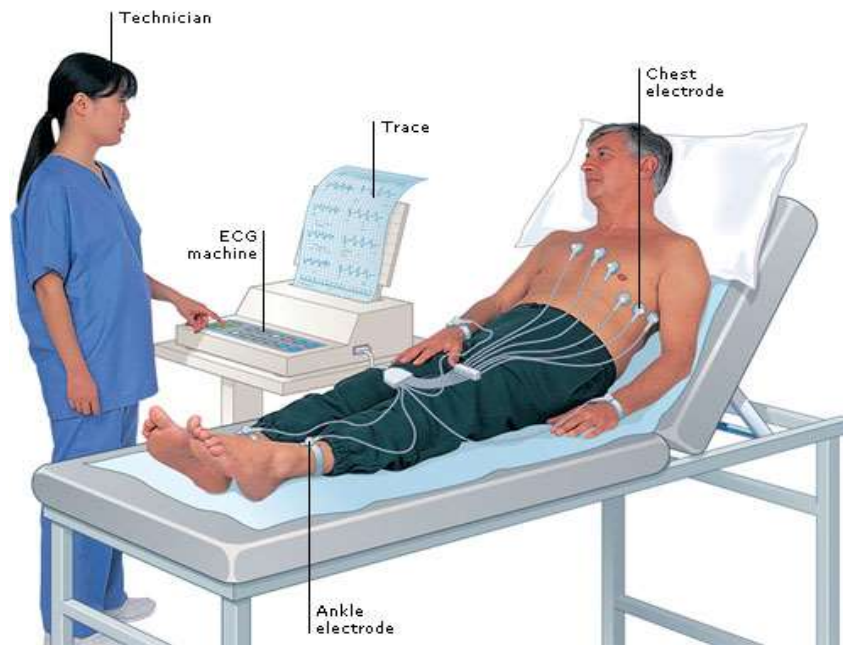


Fig 3.1 ECG Recording

During each heartbeat, a healthy heart has an orderly progression of depolarization that starts with pacemaker cells in the sinoatrial node, spreads throughout the atrium, and passes through the atrioventricular node down into the bundle of His and into the Purkinje fibres, spreading down and to the left throughout the ventricles. This orderly pattern of depolarization gives rise to the characteristic ECG tracing. To the trained clinician, an ECG conveys a large amount of information about the structure of the heart and the function of its electrical conduction system. Among other things, an ECG can be used to measure the rate and rhythm of heartbeats, the size and position of the heart chambers, the presence of any damage to the heart's muscle cells or conduction system, the effects of heart drugs, and the function of implanted pacemakers.

3.1.1 P Wave

The P wave is a summation wave generated by the depolarization front as it transits the atria. Normally the right atrium depolarizes slightly earlier than left atrium since the depolarization wave originates in the sinoatrial node, in the high right atrium and then travels to and through the left atrium. The depolarization front is carried through the atria

along semi-specialized conduction pathways including Bachmann's bundle resulting in uniform shaped waves. Depolarization originating elsewhere in the atria (atrial ectopic) result in P waves with a different morphology from normal.



Fig 3.2 : P Wave

- Peaked P waves (> 0.25 mV) suggest right atrial enlargement, cor pulmonale, (*P* pulmonale rhythm), but have a low predictive value ($\sim 20\%$).
- A P wave with increased amplitude can indicate hypokalaemia. It can also indicate right atrial enlargement.
- A P wave with decreased amplitude can indicate hyperkalaemia.
- P-wave changes in left and right atrial hypertrophy
- Bifid P waves (known as *P* mitrale) indicate left-atrial abnormality - e.g. dilatation or hypertrophy.

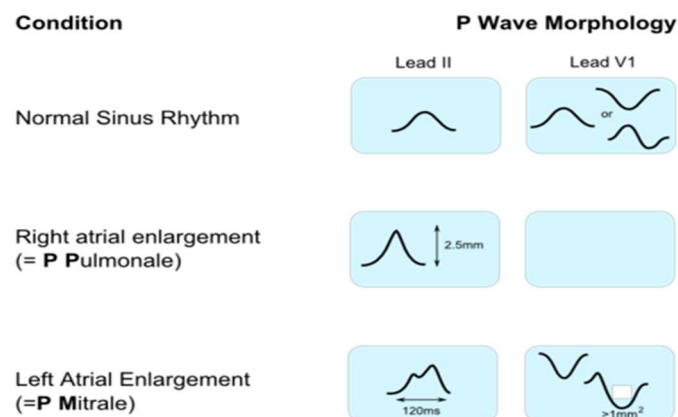


Fig 3.3 : P wave Morphology

If at least three different shaped P waves can be seen in a given ECG lead tracing, this implies that even if one of them arises from the SA node, at least two others are arising elsewhere. This is taken as evidence of multiple (i.e. at least two) ectopic foci, and is called multifocal (or more correctly, multiform) atrial rhythm if the rate is ≤ 100 or multifocal atrial tachycardia if the rate is over 100. This appears particularly commonly in exacerbations of chronic obstructive lung disease.

If the baseline has a totally irregular form, this suggests fibrillatory waves of atrial fibrillation or possibly artefact; a saw tooth shaped baseline suggests the flutter waves of atrial flutter. With either of these rhythms, if the ventricular rate is fast, the fibrillatory or flutter waves can easily be misinterpreted as P waves.

Absence of the P wave with a flat baseline may indicate:

- Fine atrial fibrillation
- Sinoatrial arrest (with a secondary escape rhythm)
- If P waves are not clearly delineated in the surface ECG, a Lewis's lead may be used to better visualize P waves.

3.1.2 QRS Complex

The QRS complex is the combination of three of the graphical deflections seen on a typical electrocardiogram (ECG or EKG). It is usually the central and most visually obvious part of the tracing. It corresponds to the depolarization of the right and left ventricles of the heart and contraction of the large ventricular muscles.

In adults, the QRS complex normally lasts 80 to 100 ms; in children it may be shorter. The Q, R, and S waves occur in rapid succession, do not all appear in all leads, and reflect a single event and thus are usually considered together. A Q wave is any downward deflection immediately following the P wave.

An R wave follows as an upward deflection, and the S wave is any downward deflection after the R wave. The T wave follows the S wave, and in some cases, an additional U wave follows the T wave. To measure the QRS interval start at the end of the PR interval to the end of the S wave. Normally this interval is 0.08 to 0.10 seconds.

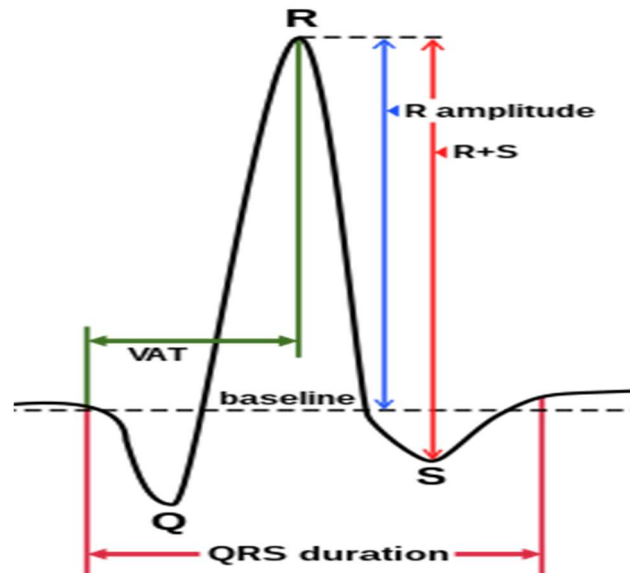


Fig 3.4 : QRS Duration

Clinical Significance :

Any abnormality of conduction takes longer and causes "widened" QRS complexes. In bundle branch block, there can be an abnormal second upward deflection within the QRS complex. In this case, such a second upward deflection is referred to as R' (pronounced "R prime"). This would be described as an RSR' pattern.

Ventricles contain more muscle mass than the atria. Therefore, the QRS complex is considerably larger than the P wave. The QRS complex is often used to determine the axis of the electrocardiogram, although it is also possible to determine a separate P wave axis.

The duration, amplitude, and morphology of the QRS complex are useful in diagnosing cardiac arrhythmias, conduction abnormalities, ventricular hypertrophy, myocardial infarction, electrolyte derangements, and other disease states. High frequency analysis of the QRS complex may be useful for detection of coronary artery disease during an exercise stress test.

ECGs help diagnose a wide range of cardiac conditions, including myocardial infarction (heart attack), arrhythmias (irregular heart rhythms), conduction abnormalities (such as bundle branch blocks), atrial and ventricular hypertrophy, pericarditis, and myocarditis. ECGs provide information about the heart's electrical conduction system and rhythm, help clinicians assess overall cardiac function.

3.1.3 T Wave:

In electrocardiography, the T wave represents the repolarization of the ventricles. The interval from the beginning of the QRS complex to the apex of the T wave is referred to as the absolute refractory period. The last half of the T wave is referred to as the relative refractory period or vulnerable period. The T wave contains more information than the QT interval. The T wave can be described by its symmetry, skewness, slope of ascending and descending limbs, amplitude and subintervals like the $T_{\text{peak}}-T_{\text{end}}$ interval.

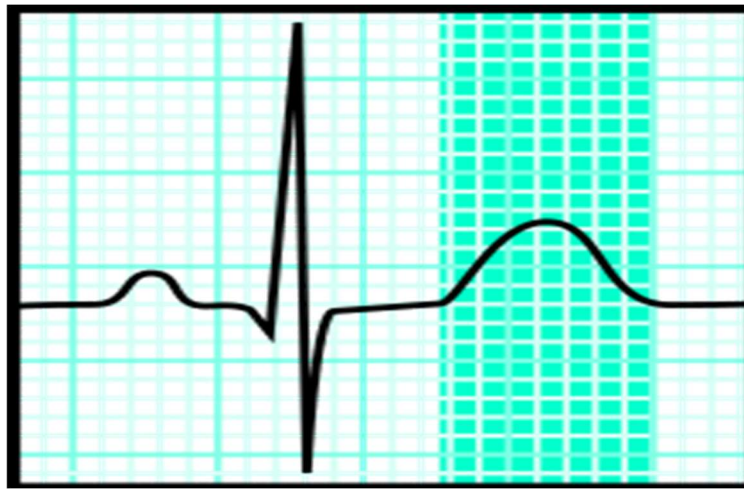


Fig 3.5 : T Wave

In most leads, the T wave is positive. This is due to the repolarization of the membrane. During ventricle contraction (QRS complex), the heart depolarizes. Repolarization of the ventricle happens in the opposite direction of depolarization and is negative current, signifying the relaxation of the cardiac muscle of the ventricles. But this negative flow causes a positive T wave; although the cell becomes more negatively charged, the net effect is in the positive direction, and the ECG reports this as a positive spike.^[2] However, a negative T wave is normal in lead aVR. Lead V1 generally have a negative T wave. In addition, it is not uncommon to have a negative T wave in lead III, aVL, or aVF. A periodic beat-to-beat variation in the amplitude or shape of the T wave may be termed T wave alternans.

Normal T Wave:

Normally, T waves are upright in all leads, except aVR and V1 leads. Highest amplitude of T wave is found at V2 and V3 leads. The shape of the T wave is usually asymmetrical with a rounded peak. T wave inversions from V2 to V4 leads are frequently found and normal in children. In normal adults, T wave inversions from V2 to V3 are less commonly found but can be normal. The depth of the T wave also becomes progressively shallow from one to the next lead. The height of the T wave should not exceed 5 mm in limb leads and more than 10 mm in precordial leads.

Inverted T wave:

Inverted T wave is considered abnormal if inversion is deeper than 1.0 mm. Inverted T waves found in leads other than the V1 to V4 leads is associated with increased cardiac deaths. Inverted T waves associated with cardiac signs and symptoms (chest pain and cardiac murmur) are highly suggestive of myocardial ischaemia. Other ECG changes associate with myocardial ischaemia are: ST segment depression with an upright T wave; ST segment depression with biphasic T wave or inverted T wave with negative QRS complex; T wave symmetrically inverted with a pointed apex, while the ST segment is either bowed upwards or horizontally depressed, or not deviated; and ST segment depression progressing to abnormal T wave during ischaemia free intervals. However, ST segment depression is not suggestive of ischaemic location of the heart. ST segment depression in eight or more leads, associated with ST segment elevation in aVR and V1 are associated with left main coronary artery disease or three-vessel disease (blockage of all three major branches of coronary arteries). ST segment depression most prominent from V1 to V3 is suggestive of posterior infarction. Furthermore, tall or wide QRS complex with an upright T wave is further suggestive of the posterior infarction.

Biphasic T wave

As the name suggests, Biphasic T waves move in opposite directions. The two main causes of these waves are myocardial ischemia and hypokalaemia.

- Schemic T waves rise and then fall below the cardiac resting membrane potential.
- Hypokalemic T waves fall and then rise above the cardiac resting membrane potential.

Wellens' Syndrome is a pattern of biphasic T waves in V2–3. It is generally present in patients with ischemic chest pain.

- Type 1: T-waves are symmetrically and deeply inverted.
- Type 2: T-waves are biphasic with negative terminal deflection and positive initial deflection.

Flattened T wave

T wave is considered flat when the wave varies from -1.0 mm to + 1.0 mm in height. Hypokalemia or digitalis therapy can cause flattened T wave with a prominent U wave. As hypokalemia progressively worsens, T wave becomes more flattened while U wave becomes more prominent, with progressively deeper ST segment depression. For digitalis toxicity, there will be sagging QT interval, flattened T wave, and prominent U wave with a shortened QT interval.

Hyperacute T Wave

These T waves may be seen in patients displaying Prinzmetal angina. Additionally, patients exhibiting the early stages of STEMI may display these broad and disproportional waves.

Camel Hump T Wave:

The name of these T waves suggests the shape it exhibits (double peaks). Since these T wave abnormalities may arise from different events, i.e. hypothermia and severe brain damage, they have been deemed as nonspecific, making them much more difficult to interpret.

Peaked T Wave:

High blood potassium levels (hyperkalemia) can cause "peaked t-waves.

3.1.4 U Wave:

The U wave is a small (0.5 mm) deflection immediately following the T wave

- U wave is usually in the same direction as the T wave.
- U wave is best seen in leads V2 and V3.

Features of Normal U waves :

- The U wave normally goes in the same direction as the T wave.

- U -wave size is inversely proportional to heart rate: the U wave grows bigger as the heart rate slows down.
- U waves generally become visible when the heart rate falls below 65 bpm.
- The voltage of the U wave is normally $< 25\%$ of the T-wave voltage: disproportionately large U waves are abnormal.
- Maximum normal amplitude of the U wave is 1-2 mm.

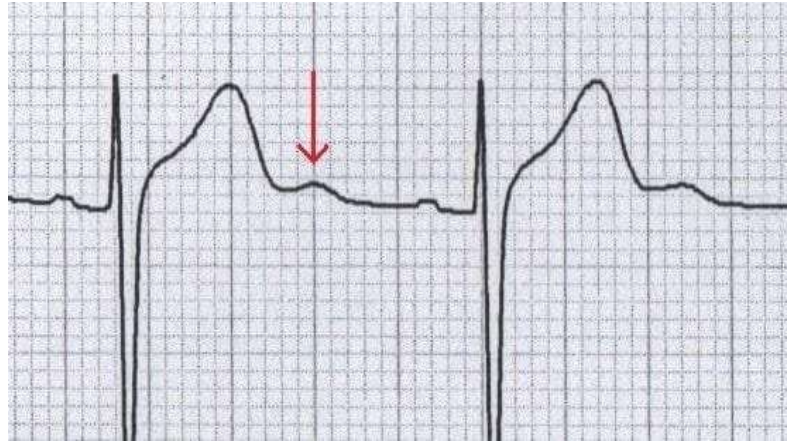


Fig 3.6 : U Wave

3.2 Electro-Cardiograph Machines:

Electrocardiograms are recorded by machines that consists of a set of electrodes connected to a central unit. Early ECG machines were constructed with analog electronics, where the signal drove a motor to print out the signal onto paper. Today, electrocardiographs use analog-to-digital converters to convert the electrical activity of the heart to a digital signal.



Fig 3.6: An ECG electrode

Many ECG machines are now portable and commonly include a screen, keyboard, and printer on a small wheeled cart. Recent advancements in electrocardiography include developing even smaller devices for inclusion in fitness trackers and smart watches. These smaller devices often rely on only two electrodes to deliver a single lead I. Portable twelve-lead devices powered by batteries are also available.

Recording an ECG is a safe and painless procedure. The machines are powered by mains power but they are designed with several safety features including an earthed (ground) lead. Other features include:

- Defibrillation protection: any ECG used in healthcare may be attached to a person who requires defibrillation and the ECG needs to protect itself from this source of energy.
- Electrostatic discharge is similar to defibrillation discharge and requires voltage protection up to 18,000 volts.
- Additionally, circuitry called the right leg driver can be used to reduce common-mode interference (typically the 50 or 60 Hz mains power).
- ECG voltages measured across the body are very small. This low voltage necessitates a low noise circuit, instrumentation amplifiers, and electromagnetic shielding.
- Simultaneous lead recordings: earlier designs recorded each lead sequentially, but current models record multiple leads simultaneously.

Most modern ECG machines include automated interpretation algorithms. This analysis calculates features such as the PR interval, QT interval, corrected QT (QTc) interval, PR axis, QRS axis, rhythm and more. The results from these automated algorithms are considered "preliminary" until verified and/or modified by expert interpretation.

3.2.1 Types of ECG Machines:

Cardiopulmonary exercise test (CPET):

A cardiopulmonary exercise test (CPET) is an evaluation of the cardiopulmonary system. This test is used to detect any cardiac or pulmonary diseases. Learn more about a cardiopulmonary exercise test.



Fig 3.7 CPET Recording

Exercise ECG (stress test):

A stress test (also called treadmill test or exercise ECG) is given while a patient walks on a treadmill or pedals a stationary bicycle to monitor the heart during stress or exercise. Breathing and blood pressure rates are also monitored. A stress test may be used to detect coronary artery disease, and/or to determine safe levels of exercise following a heart attack or heart surgery. Learn more about a stress test.

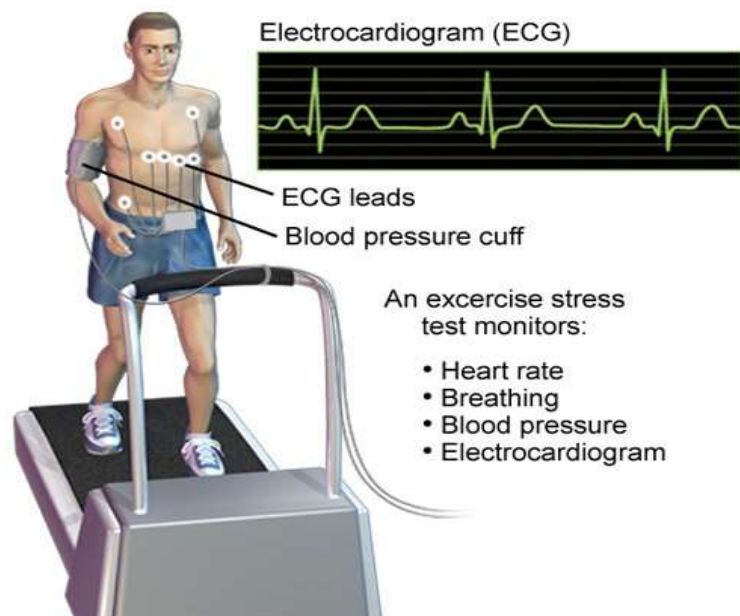


Fig 3.8: Stress Test Recording

Holter monitor:

The Holter monitor is a type of electrocardiogram (ECG or EKG) used to monitor the ECG tracing continuously for a period of 24 hours or longer. Electrodes (small, plastic patches) are placed at certain locations on the chest, arms, and legs. When the electrodes are connected to an ECG machine by lead wires, the electrical activity of the heart is measured, interpreted, and printed out for the physician's information and further interpretation. Learn more about a Holter monitor.

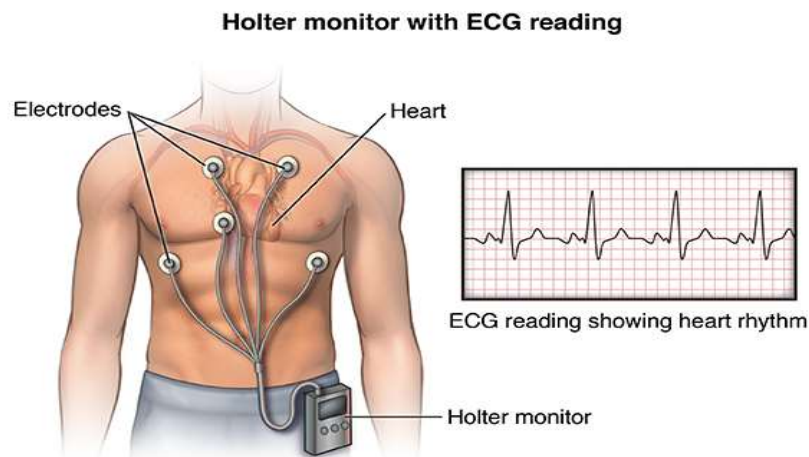


Fig 3.9 : Holter Monitor Recording

Resting 12-lead ECG:

This type of electrocardiogram is the standard test for measuring your heart's electrical function. Performed while you are lying still, this ECG records your heart's electrical activity from 12 electrodes (sticky patches) on your chest, arms, and legs at the same time. A resting 12-lead ECG can be part of a routine checkup to screen for heart conditions before any signs or symptoms develop.

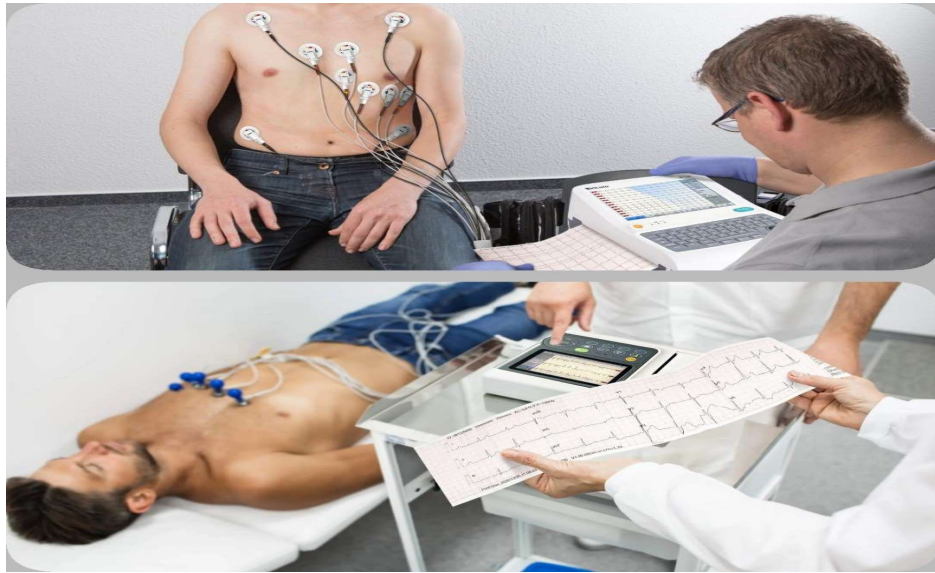


Fig 3.10 : Resting 12-lead ECG Recording

Signal-averaged electrocardiogram:

A signal-averaged electrocardiogram is a more detailed type of electrocardiogram (ECG or EKG). During this procedure, multiple ECG tracings are obtained over a period of approximately 20 minutes in order to capture abnormal heartbeats which may occur only intermittently. Learn more about a signal-averaged electrocardiogram.

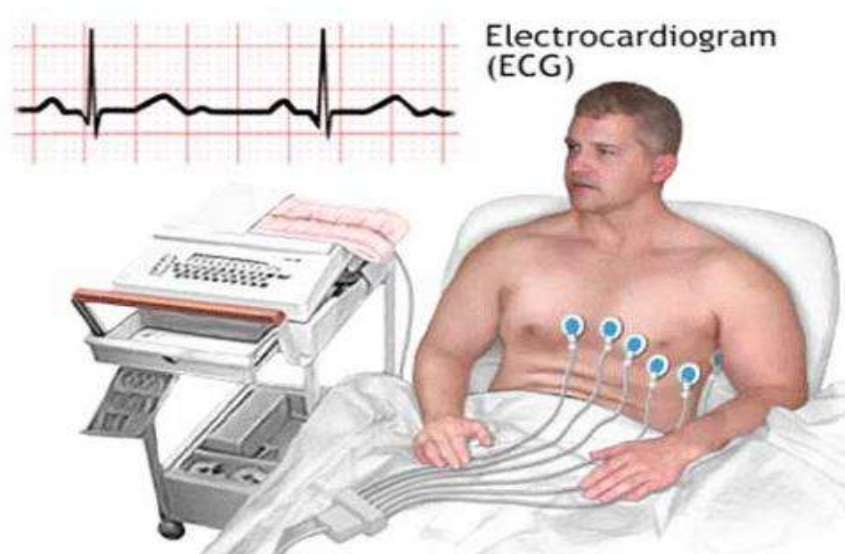


Fig 3.11 : Signal-averaged ECG Recording

3.3 Medical Uses

The overall goal of performing an ECG is to obtain information about the electrical functioning of the heart. Medical uses for this information are varied and often need to be combined with knowledge of the structure of the heart and physical examination signs to be interpreted. Some indications for performing an ECG include the following:

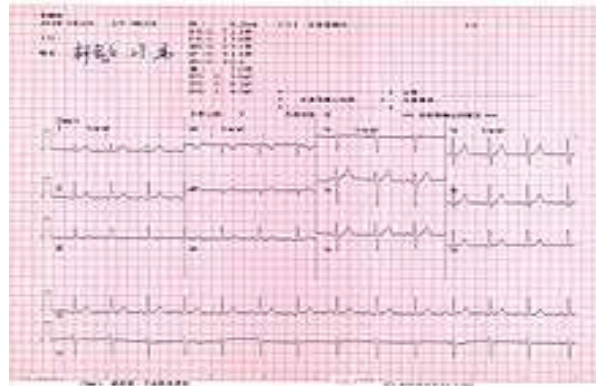


Fig 3.12 : Normal 12-lead ECG

- Chest pain or suspected myocardial infarction (heart attack), such as ST elevated myocardial infarction (STEMI) or non-ST elevated myocardial infarction (NSTEMI).
- Symptoms such as shortness of breath, murmurs, fainting, seizures, funny turns, or arrhythmias including new onset palpitations or monitoring of known cardiac arrhythmias.
- Medication monitoring (e.g., drug-induced QT prolongation, Digoxin toxicity) and management of overdose (e.g., tricyclic overdose).
- Electrolyte abnormalities, such as hyperkalemia.
- Perioperative monitoring in which any form of anesthesia is involved (e.g., monitored anesthesia care, general anesthesia). This includes preoperative assessment and intraoperative and postoperative monitoring.
- Cardiac stress testing
- Computed Tomography Angiography (CTA) and Magnetic Resonance Angiography (MRA) of the heart (ECG is used to “gate” the scanning so that the anatomical position of the heart is steady).

- Clinical cardiac electrophysiology, in which a catheter is inserted through the femoral vein and can have several electrodes along its length to record the direction of electrical activity from within the heart.

ECGs can be recorded as short intermittent tracings or continuous ECG monitoring. Continuous monitoring is used for critically ill patients, patients undergoing general anaesthesia, and patients who have an infrequently occurring cardiac arrhythmia that would unlikely be seen on a conventional ten-second ECG. Continuous monitoring can be conducted by using Holter monitors, internal and external defibrillators and pacemakers, and/or biotelemetry.

CHAPTER-4

PROPOSED METHOD

4.1 Introduction

Electrocardiogram (ECG) signals, which represent the electrical activity of the heart, can be susceptible to various types of noise that may distort the underlying cardiac information. Common types of noise that can corrupt ECG signals include baseline wander, power line interference (often at 50 or 60 Hz depending on the region), muscle noise, electrode motion artifacts, and electromagnetic interference, white gaussian noise, salt and pepper noise, random noise.

The effect of adding noise to an ECG signal can be detrimental as it obscures the true cardiac activity, making it challenging for healthcare professionals to accurately interpret the signal. Noise can distort the shape, timing, and amplitude of ECG waves such as the P-wave, QRS complex, and T-wave, which are vital for diagnosing various cardiac conditions. For example, the presence of baseline wander can shift the baseline of the ECG signal, making it difficult to distinguish between abnormal and normal cardiac activity.

The importance of removing noise from ECG signals cannot be overstated. Denoising techniques play a crucial role in enhancing the diagnostic utility of ECG signals by restoring the fidelity of the cardiac information. By effectively removing noise, denoising algorithms help improve the accuracy of ECG interpretation, leading to more reliable diagnoses and better patient outcomes. Moreover, denoising is essential for ensuring the integrity of ECG signals during acquisition and transmission, particularly in settings prone to interference such as hospitals or ambulances.

Several methods exist for removing noise from ECG signals, ranging from simple filtering techniques to advanced signal processing algorithms. Filtering approaches, such as finite impulse response (FIR) or infinite impulse response (IIR) filters, can be effective for attenuating specific frequency bands associated with noise sources like power line interference. Adaptive filters, which dynamically adjust their parameters based on the characteristics of the input signal, offer enhanced noise suppression capabilities, particularly in situations where noise characteristics may vary over time.

4.2 Empirical Mode Decomposition

Empirical Mode Decomposition (EMD) is a powerful signal processing technique that can be particularly effective for denoising ECG signals due to its adaptability, ability to capture non-stationary and nonlinear components, and inherent data-driven nature.

1. Adaptability to Signal Characteristics :

ECG signals are inherently non-stationary and can exhibit complex, time-varying dynamics. Traditional Fourier-based methods like Fourier transform or wavelet transform may not adequately capture these non-stationary components. EMD, however, is specifically designed to handle non-stationary signals and can adaptively decompose them into a set of intrinsic mode functions (IMFs) that represent different scales or oscillatory modes present in the signal. This adaptability makes EMD well-suited for capturing the dynamic nature of ECG signals, which can contain various morphologies and frequencies.

2. Preservation of Signal Features :

One of the key advantages of EMD is its ability to preserve important signal features while removing noise. During the decomposition process, EMD separates the signal into IMFs based on local extrema and zero-crossings, effectively capturing both high-frequency noise and low-frequency signal components in different IMFs. Since each IMF represents a distinct oscillatory mode present in the signal, denoising can be achieved by selectively reconstructing the signal using only the IMFs corresponding to the relevant cardiac information while discarding those associated with noise.

3. Data Driven Approach :

EMD does not rely on predefined basis functions or filter kernels, unlike traditional Fourier-based or wavelet-based methods. Instead, it is entirely data-driven, making minimal assumptions about the underlying signal structure. This data-adaptive nature allows EMD to effectively adapt to the unique characteristics of each ECG signal, making it robust against variations in noise levels, signal morphology, and interference sources. As a result, EMD-based denoising can be more flexible and versatile, capable of handling diverse ECG recordings encountered in clinical practice.

4. Nonlinear Signal Processing :

EMD operates based on the concept of sifting through the signal's data points to extract

oscillatory modes, making it inherently nonlinear. This nonlinearity enables EMD to capture complex signal dynamics and morphologies that may be challenging for linear denoising techniques. In the context of ECG signals, which can exhibit nonlinear phenomena such as arrhythmias or irregular heartbeats, EMD's ability to capture nonlinear features can be advantageous for preserving the integrity of the cardiac information while removing noise.

4.3 Flow of EMD Method

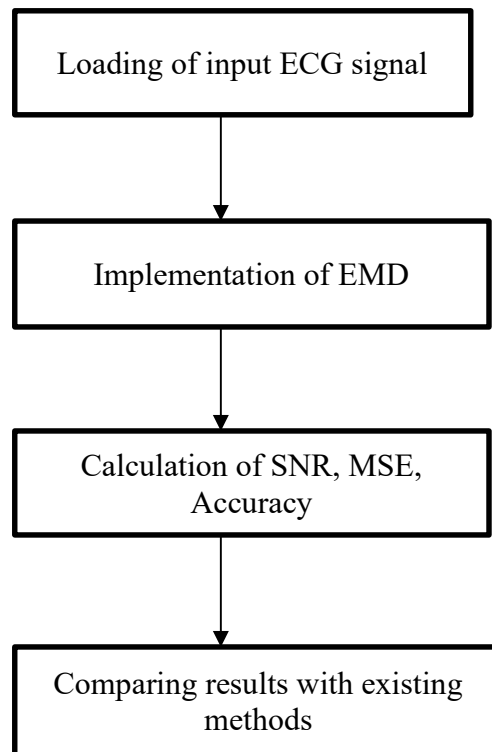


Fig 4.1 : Flow of EMD method

1.Loading of Input ECG Signal:

Loading an ECG signal into MATLAB involves several steps, beginning with obtaining the ECG data in a compatible format and then using MATLAB's built-in functions or custom scripts to import and process the signal.

- **Data Acquisition:**

ECG data can be obtained from various sources, such as physiological recording devices, databases, or stored files. Ensure that the ECG data is available in a format that

MATLAB can recognize, such as plain text files (e.g., CSV or TXT), MATLAB data files (MAT-files), or specific biomedical data formats like PhysioNet's WFDB format.

- **File Preparation:**

Before loading the ECG signal into MATLAB, it's essential to ensure that the data file is properly formatted and organized. For example, if the ECG data is stored in a text file, make sure it follows a consistent structure with appropriate headers (if any) and column labels indicating the time and voltage values of the ECG signal.

- **Using MATLAB Functions:**

MATLAB provides several built-in functions for reading data files into the workspace. For example, the **load** function can be used to load data from MAT-files, while functions like **'csvread'** or **'readtable'** can be used to import data from CSV or text files, respectively. If the ECG data is stored in a proprietary or specialized format, MATLAB may offer specific functions or toolboxes (e.g., the PhysioNet toolbox) to facilitate data import.

- **Data Visualization:**

Once the ECG signal is loaded into MATLAB, it's often helpful to visualize the data to ensure it was imported correctly and to get an overview of its characteristics. MATLAB's plotting functions, such as **plot** or **plotly**, can be used to create time-domain plots of the ECG waveform, with time on the x-axis and voltage on the y-axis.

- **Signal Processing :**

Depending on the analysis or processing tasks, additional steps may be required to preprocess or analyze the ECG signal. This may include filtering to remove noise, detecting QRS complexes, calculating heart rate, or extracting specific features for further analysis. MATLAB provides a comprehensive set of signal processing functions and toolboxes (e.g., Signal Processing Toolbox, Wavelet Toolbox) to perform these tasks efficiently.

- **Saving Results:**

After processing or analyzing the ECG signal, you may want to save the results for further use or documentation. MATLAB's **save** function can be used to save variables or data structures to MAT-files, while other functions like **writeable** or **csvread** can be used to export data to CSV or text files.

2. Implementation of EMD:

The process of Empirical Mode Decomposition (EMD) involves several steps aimed at decomposing a signal into a set of intrinsic mode functions (IMFs) that represent different scales or oscillatory modes present in the signal. Here's a detailed explanation of the EMD process:

- **Signal Decomposition:**

Begin by decomposing the original signal $x(t)$ into a finite number of IMFs, $c_1(t)$, $c_2(t)$, $c_n(t)$, and a residual signal $r(t)$.

The decomposition can be represented as:

$$x(t) = \sum_{i=1}^n c_i(t) + r(t) \quad 4.1$$

Each IMF represents a distinct oscillatory component of the signal, ranging from high-frequency components to low-frequency trends.

- **Shifting Process :**

Perform the sifting process iteratively to extract IMFs from the original signal. The sifting process involves the following steps:

- Identify the local extrema (maxima and minima) in the signal.
- Interpolate between consecutive maxima and minima to obtain the upper envelope (maxima) and lower envelope (minima).
- Calculate the mean of the upper and lower envelopes to obtain the local mean.
- Subtract the local mean from the original signal to obtain the first IMF.
- Repeat the process with the obtained IMF as the new input signal until certain stopping criteria are met. For an instance, when the residual becomes a monotonic function.

- **Stopping Criterion:**

Define stopping criteria for the sifting process to determine when to terminate the decomposition. Common stopping criteria include:

Monotonically: Stop when residual becomes a monotonic function.

Extrema Limit: Stop when the number of extrema in the residual falls below a certain threshold.

Iteration Limit: Stop after a predefined number of iterations to avoid excessive

computational time.

- **Residual Handling:**

Handle the residual obtained after extracting the desired number of IMFs. The residual represents the trend component of the original signal and may require further processing or analysis depending on the application.

3.Calculation of Metrics:

Calculating Signal-to-Noise Ratio (SNR), Mean Squared Error (MSE), and accuracy are important metrics used in signal processing and machine learning applications to assess the quality of signal processing algorithms, denoising techniques, or classification models.

1.Signal-to-Noise Ratio (SNR) :

SNR quantifies the level of desired signal present in relation to background noise or interference. A higher SNR indicates a better-quality signal with less noise contamination, which is crucial for accurate signal interpretation and analysis.

SNR is typically calculated as the ratio of the power of the signal to the power of the noise. In decibels (dB), it is often expressed as

$$SNR_{dB} = 10 \log_{10} \left(\frac{\text{Signal Power}}{\text{Noise Power}} \right) \quad 4.2$$

$$\text{Where signal power} = \frac{1}{N} \sum_{i=1}^N x_i^2$$

$$\text{Noise power} = \frac{1}{N} \sum_{i=1}^N n_i^2$$

2.Mean Square Error(MSE) :

MSE provides a quantitative measure of the accuracy of an estimation or prediction algorithm. Lower MSE values indicate better accuracy and closer agreement between the estimated and true values. MSE measures the average squared difference between the estimated values and the true values. It is calculated as

$$MSE = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2 \quad 4.3$$

where Y_i = true value of y

$\hat{Y}_i$ = estimated value of y

3.Accuracy:

Accuracy typically refers to the fidelity of the decomposition process in capturing the intrinsic oscillatory modes or components present in a signal. Since EMD is a data-driven method without predefined basis functions, its accuracy is often assessed based on how well it decomposes a signal into intrinsic mode functions (IMFs) that represent the signal's local oscillatory behavior.

$$\text{RMSE} = \sqrt{\text{mean}((x(1,:) - \text{recsig}).^2)} \quad 4.4$$

where x = input signal

recsig = reconstructed signal

$$\text{Accuracy} = 100 * (1 - \text{RMSE} / (\max(x) - \min(x))) \quad 4.5$$

These metrics provide objective measures to evaluate the performance of signal processing algorithms, denoising techniques, or machine learning models. By quantifying the quality of signal processing or classification results, these metrics help researchers and practitioners compare different approaches and select the most effective ones for their specific applications. Calculating SNR, MSE, and accuracy allows for quality assurance in signal processing. By monitoring these metrics, practitioners can ensure that their algorithms or models meet predefined performance criteria and deliver reliable results in real-world scenarios.

CHAPTER – 5

IMPLEMENTATION

5.1 Methodology :

In this we are going to explain the step-by-step process of the project with the help of a block diagram shown in Figure-5.1. It consists of extracting features and denoising using proposed method and then existing method to compare the results.

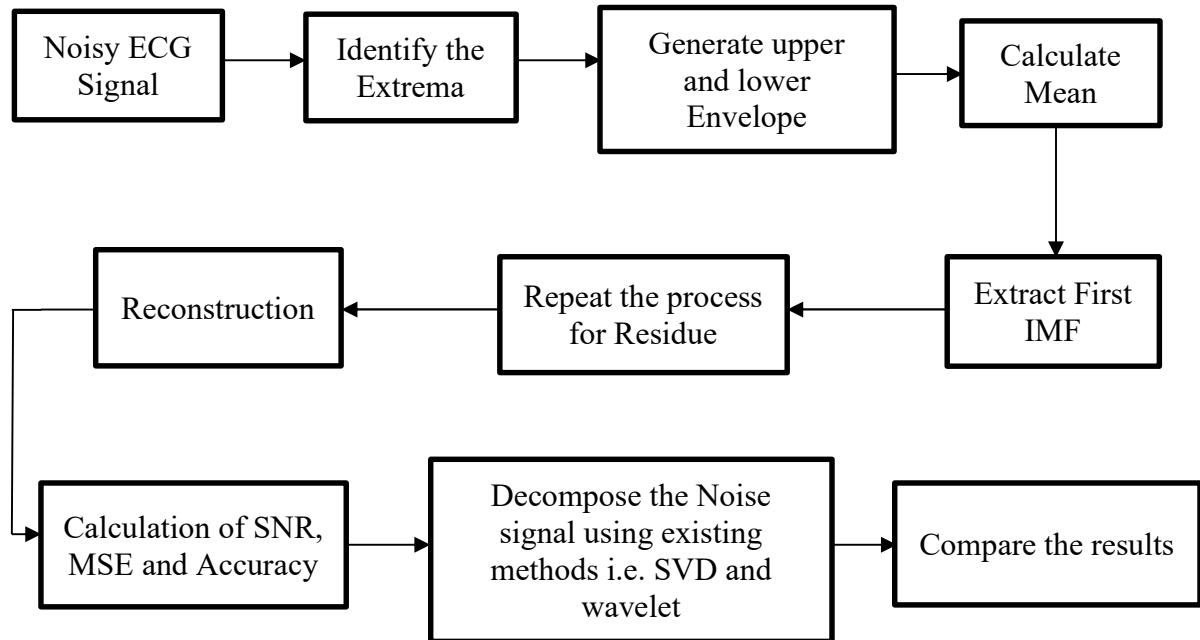


Fig 5.1 : Block Diagram of steps involved in this project

1.Noisy ECG Signal :

Empirical Mode Decomposition starts with the raw signal data. Ensure that the data is non-linear and non-stationary, meaning its statistical properties change over time. The input signal is downloaded from physio Net website with .mat extension and load the function using 'load' from the selected file into a MATLAB structure. By using 'uigetfile' function, open a dialog box for selected files. It allows user to navigate through their file system and choose one or more files. It allows user to navigate through their file system and choose one or more files. Then add different types of noise to analysis the robustness of the algorithm and evaluation of the algorithm.

2. Identify the extrema:

Identifying extrema in an ECG signal involves locating points of maximum and minimum amplitude, which correspond to peaks and valleys in the waveform. One common method is to apply peak detection algorithms to the ECG signal. These algorithms search for local maxima and minima by comparing each data point with its neighboring points. Peaks are identified as local maxima that exceed a certain threshold, while valleys are identified as local minima that fall below a specified threshold. Various techniques, such as thresholding, derivative-based methods, or wavelet transforms, can be employed for peak detection in ECG signals. Once extrema are detected, they serve as key landmarks for analyzing cardiac activity, including measuring heart rate, identifying abnormal rhythms, and extracting features for diagnostic purposes.

3. Generate Upper and lower envelope:

To generate the upper and lower envelope of an ECG signal, start by identifying local maxima and minima, which represent points where the signal changes direction. Then, interpolate between adjacent maxima to create the upper envelope and between adjacent minima to create the lower envelope. This interpolation process smooths out the envelope curves, ensuring they closely follow the variations in the signal. Finally, plot the original ECG signal along with the upper and lower envelope curves to visualize the signal's dynamics and amplitude variations.

4. Calculate mean:

Compute the mean of the upper and lower envelopes. This mean serves as a reference for the local oscillations.

5. Extract the first IMF:

Subtract the mean from the original signal to obtain the first component, which represents the local oscillations at the shortest timescale. Repeat the process on the obtained component until certain stopping criteria are met. These criteria usually assess the amplitude of the residual and the number of extrema.

$$\epsilon = \left(\sum \text{IMF}_i - \sum \text{IMF}_{i-1} \right) \div \sum \text{IMF}_i^2 \quad 5.1$$

where ϵ = epsilon lies between 0 and 1

Check if the obtained component satisfies the above the condition of being IMF. An IMF should have the same number of zero crossings and extrema, or differ by at most one. If the obtained component does not satisfy these conditions, it is not considered an IMF and the process is repeated on the residual which is the original signal minus the obtained component).

6.Repeat the process for residual:

Apply the EMD process iteratively on the residual until the stopping criterion is met. Each iteration yields an IMF. The final residual is the trend component of the signal, which is usually a slowly varying component representing the overall trend of the data. Stop the process when either the residual becomes a monotonic function or satisfies a certain threshold level of amplitude. Once all IMFs and residual are obtained, they can be analyzed individually or combined to reconstruct the original signal.

7.Reconstruction:

Reconstruction in Empirical Mode Decomposition (EMD) is the process of reconstructing the original signal from its Intrinsic Mode Functions (IMFs) and residual. After decomposing the signal into IMFs through the EMD algorithm, the reconstructed signal is obtained by summing the IMFs and the residual component. Each IMF represents a component of the original signal at a specific timescale or frequency, capturing its oscillatory behavior. The residual component represents the trend or low-frequency content of the signal that remains after extracting all IMFs. Reconstruction allows for the analysis and visualization of the original signal's components, aiding in the interpretation of its underlying dynamics and features. It is an essential step in EMD-based signal processing applications such as denoising, feature extraction, and trend analysis.

8.Calculation of SNR, MSE and Accuracy

Signal-to-Noise Ratio (SNR), Mean Squared Error (MSE), and Accuracy are common metrics used to assess the performance of signal processing algorithms, including those based on Empirical Mode Decomposition (EMD). Here's a brief explanation of each metric and why they are important:

1. Signal-to-Noise Ratio (SNR):

- SNR measures the ratio of signal power to noise power in a signal. It quantifies how much the signal power exceeds the noise power, providing an indication of the signal's quality relative to the noise level.
- High SNR values indicate that the signal is dominant compared to the noise, while low SNR values suggest that the noise is significant relative to the signal.
- SNR is crucial for assessing the fidelity of signal processing algorithms, such as denoising techniques. A higher SNR indicates better noise suppression and preservation of signal content.

2. Mean Squared Error (MSE):

- MSE quantifies the average squared difference between the estimated signal (reconstructed signal) and the true signal (original signal). It measures the overall accuracy of the estimation or reconstruction process.
- Lower MSE values indicate smaller errors between the estimated and true signals, reflecting better accuracy and fidelity of the reconstruction.
- MSE is commonly used as a loss function in optimization algorithms for signal processing tasks, where the goal is to minimize the discrepancy between the estimated and true signals.

3. Accuracy:

- Accuracy measures the proportion of correctly classified or reconstructed data points relative to the total number of data points. It assesses the correctness of the algorithm's output compared to ground truth or reference data.
- High accuracy values indicate a high level of agreement between the estimated and true signals, while low accuracy values suggest discrepancies or errors in the reconstruction process.
- Accuracy is particularly important in classification tasks or when comparing the performance of different algorithms in terms of their ability to accurately classify or reconstruct signals.

These metrics play a vital role in evaluating the performance of signal processing algorithms, guiding algorithm design, parameter tuning, and model selection. They provide

quantitative measures of the algorithm's effectiveness in preserving signal content, suppressing noise, and accurately reconstructing the original signal, thereby facilitating informed decision-making and algorithm optimization.

9. Decompose the Signal using existing methods:

1. Decomposition using Singular Value Decomposition (SVD):

- Singular Value Decomposition is a matrix factorization technique that decomposes a matrix into three matrices: U , Σ , and V . For signal decomposition, the data matrix is constructed by organizing the signal samples into columns or rows.
- SVD decomposes the data matrix into three matrices: U , Σ , and V . U and V are orthogonal matrices containing left and right singular vectors, respectively. Σ is a diagonal matrix containing singular values.
- To decompose a signal using SVD, construct the data matrix from the signal samples, perform SVD on the data matrix, and select a subset of singular vectors and singular values to reconstruct the signal.
- SVD-based decomposition can be useful for analyzing the dominant modes or components of a signal and for denoising applications.

2. Decomposition using Wavelet Transform:

- Wavelet Transform is a mathematical transform that decomposes a signal into wavelet coefficients representing different frequency components at various scales.

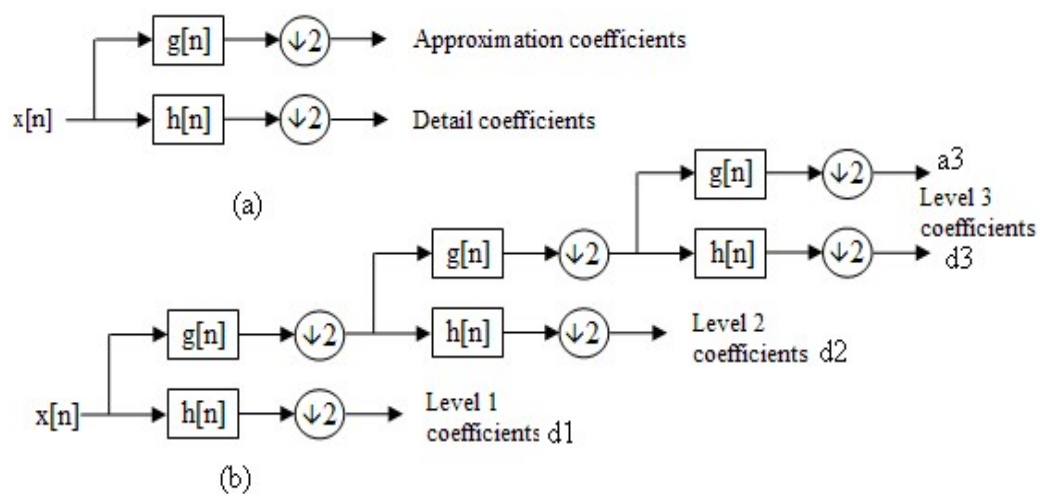


Fig 5.2 : (a) Wavelet Decomposition for level1 (b) Wavelet Decomposition for level 3

- Wavelet transform operates by convolving the signal with wavelet functions at different scales and positions. It generates a multi-resolution analysis of the signal, capturing both high and low-frequency components.
- The wavelet transform produces a set of coefficients representing the signal's approximation and detail at each decomposition level.
- To decompose a signal using wavelet transform, select a wavelet basis function (e.g., Haar, Daubechies), choose the decomposition level, and apply the wavelet transform to the signal.
- Wavelet-based decomposition is widely used for feature extraction, denoising, compression, and signal analysis tasks due to its ability to capture signal characteristics at different scales.

10. Compare the results:

Compare the results obtained by extracting features and denoising using Empirical Mode Decomposition and existing methods Singular Value Decomposition and wavelet Decomposition.

CHAPTER-6

RESULTS AND DISCUSSIONS

In this section, we are going to discuss about the results from the execution of the code. The test ECG signals are taken from the physio Net ATM and based time and amplitude the features of the ECG signal are extracted and to analysis the EMD algorithm different types of noises like white Gaussian Noise, Random Noise, Salt and Pepper Noise, Power Line Interference Noise and Baseline Wander Noise to analyze the difference between existing algorithm and proposed algorithm. For analysis, the metrics SNR, MSE and accuracy is calculated and waveforms are plotted.

Below are waveforms obtained when the random noise is added to 102m.mat(sample2) signal and undergoes SVD, Wavelet and EMD and SNR, MSE and Accuracy is calculated and observed better accuracy for this test sample.

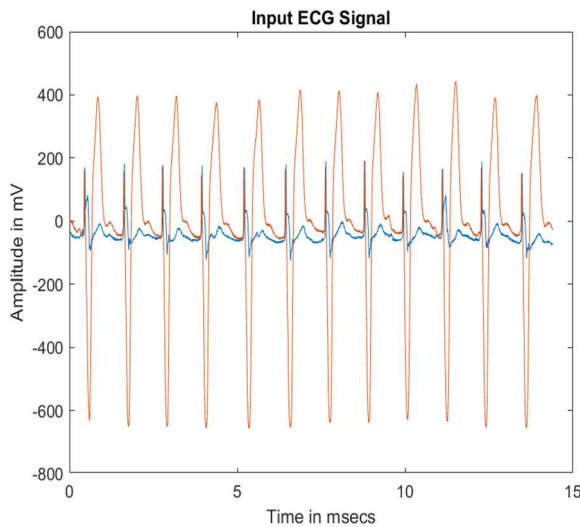


Fig 6.1 : Input ECG Signal

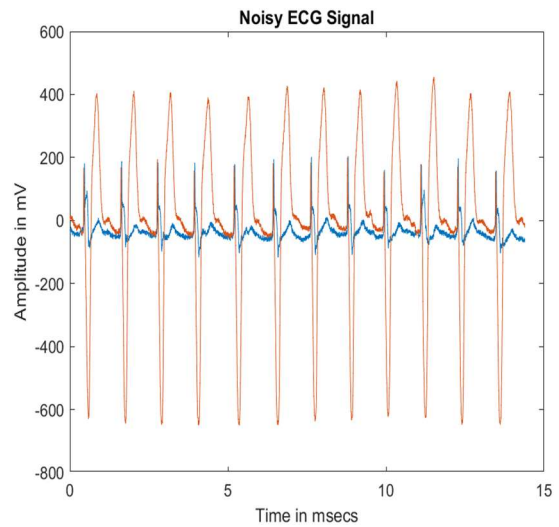


Fig 6.2 : Noisy ECG Signal

The Input signal taken from Physio Net with .mat extension and the code begin by importing an ECG signal from a file selected by the user using 'uigetfile' function and plot the signal and random noise is added to the input signal for the evaluation of algorithm and noisy signal is plotted for visualization.

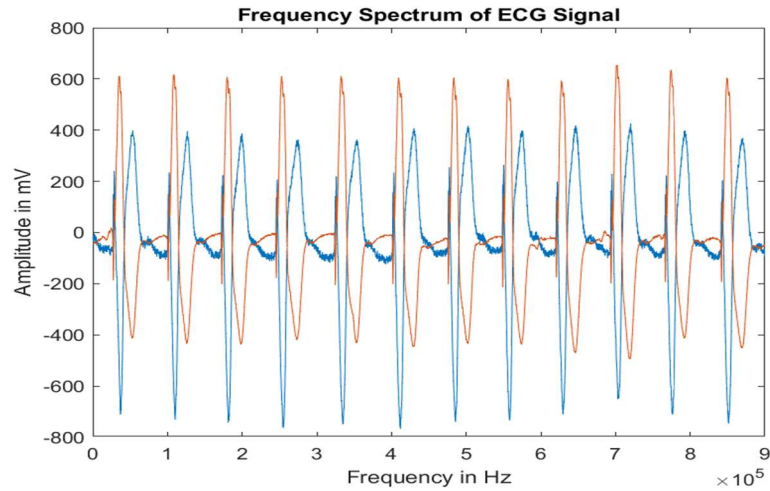


Fig 6.3 : Frequency Spectrum of ECG Signal

The Fast Fourier Transform(FFT) is applied to the noisy ECG signal to obtain its frequency spectrum. The frequency spectrum is plotted to visualize the signal's frequency components. The FFT provides a way to transform a signal from the time domain to the frequency domain. By applying the FFT to the noisy ECG signal, we can examine the distribution of frequency components present in the signal. Noisy signals often contain unwanted frequency components that can distort or mask the underlying ECG signal. By analyzing the frequency spectrum obtained from the FFT, we can identify the frequencies at which noise is most prominent.

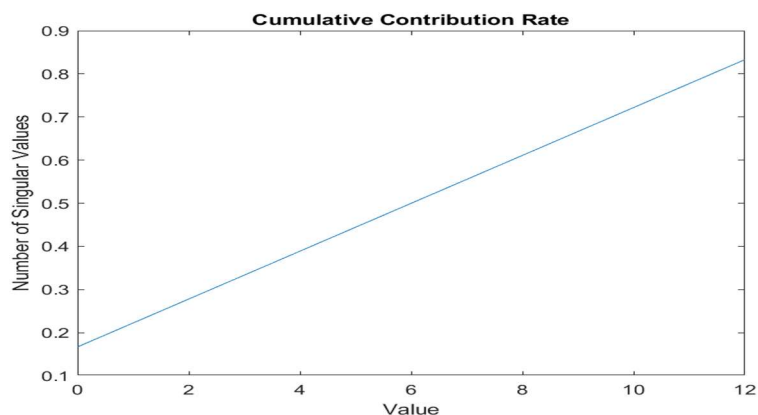


Fig 6.4 : Cumulative Contribution Rate

Singular Value Decomposition(SVD) is performed on the noisy ECG signal to decompose it into its singular values and components. The singular values and components are plotted to visualize their contributions to the signal. Fig 6.4 is cumulative contribution rate and waveform is linear which indicates all singular values holds the equal information.

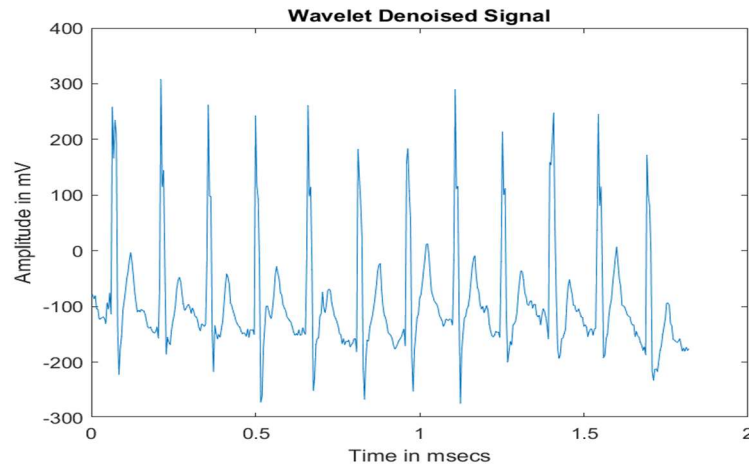


Fig 6.5 : Wavelet Denoised Signal

Discrete Wavelet Transform (DWT) is applied to the noisy ECG signal using the 'sym4' wavelet. The denoised signal is obtained from the DWT is plotted for visualization.

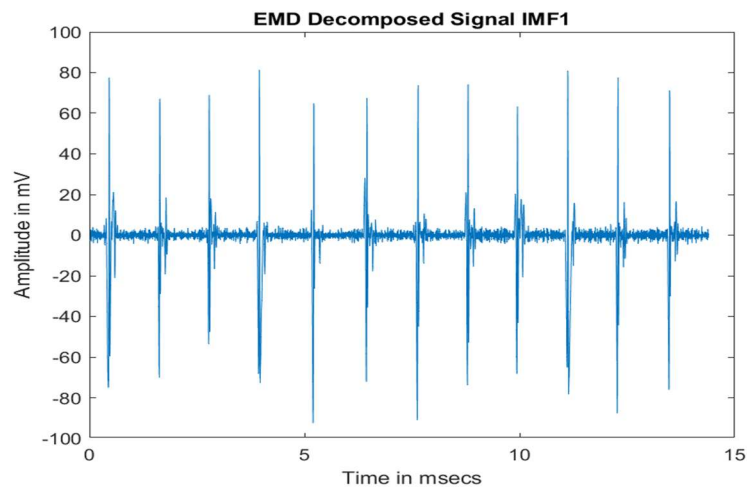


Fig 6.6 : EMD Decomposed First IMF

EMD decomposes a signal into finite number of oscillatory components called Intrinsic Mode Functions (IMFs) and a trend component known as residual. Each IMF represents a component of the signal with a characteristic frequency and time scale, capturing local oscillations or modes present in the signal. Theoretically, IMFs are extracted through an iterative process of shifting, where local extrema are identified and connected with cubic splines to obtain the first IMF but MATLAB provide inbuilt function to decompose the signal into IMFs and residual. The fig 6.6 shows the first IMF and the process is repeated to extract subsequent IMFs, each representing progressively smaller scale oscillations.

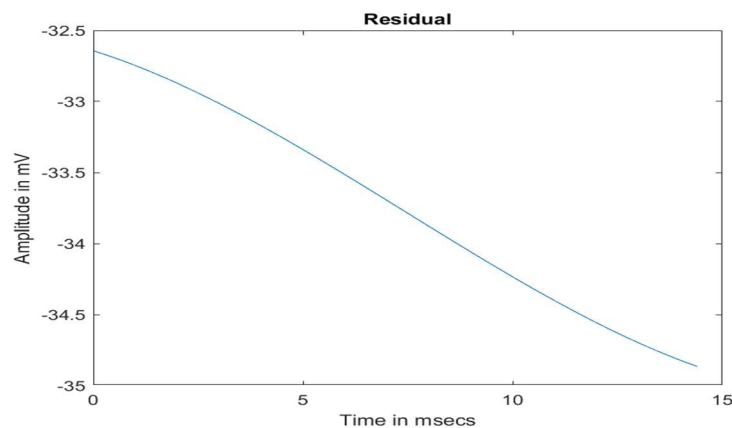


Fig 6.7 Residual Signal

The residual is the remaining signal after extracting all IMFs, representing the overall trend or low-frequency components of the signal that are not captured by the IMFs. The residual is the part of the original signal that cannot be decomposed into a finite number of IMFs using EMD process. It represents the low-frequency or trend-like components of the signal that are not captured by the IMFs. Mathematically, the residual is obtained by subtracting the sum of all IMFs from the original signal. The residual can be visualized by plotting it against time, similar to the original signal and IMFs. Plotting the residual allows for the inspection of its waveform.

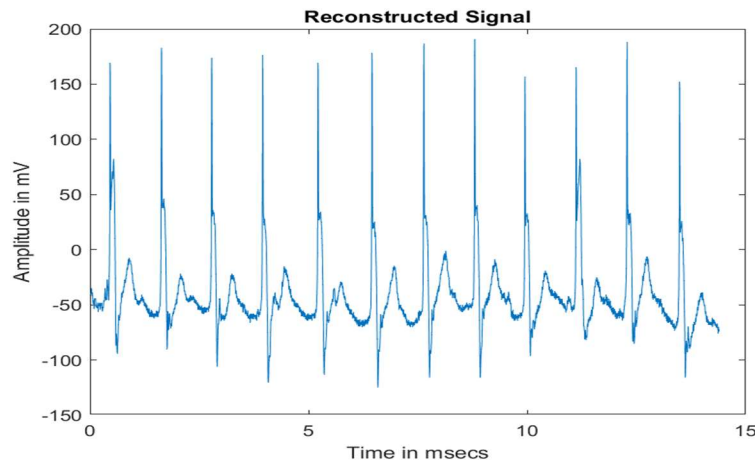


Fig 6.8 : Reconstructed Signal using EMD

The Original signal can be reconstructed by summing all IMFs and the residual. This reconstructed signal closely resembles the original signal, capturing its essential features while removing noise and undesirable components. These are the waveforms obtained when random signal is added to 102m.mat signal at minimum value is 1 and maximum is 15. The metrics are also calculated for different noises at different levels for performance evaluation.

Signal-to-Noise Ratio (SNR) and Mean Square Error(MSE) are calculated to evaluate the performance of the denoising techniques (Wavelet, EMD). SNR and MSE are calculated for each denoised signal. Additionally, accuracy is computed as a percentage based on the Root Mean Square Error (RMSE) between the original and reconstructed signals, indicating how closely the reconstructed signal matches the original.

Based on test samples, we conclude that EMD algorithm works well for low MSE. For each test sample the MSE error is very low than SVD and wavelet for every test sample and for all the noises that we added to the signal for evaluation of algorithm.

For Salt and Pepper noise, we observe very high SNR than SVD and we achieved equal SNR for EMD and wavelet but MSE is low in EMD than wavelet. Below are the simulated waveforms for 100m.mat sample signal with salt and pepper noise.

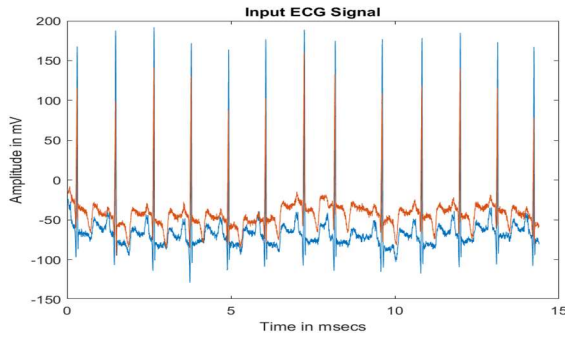


Fig 6.9 : Input ECG Signal

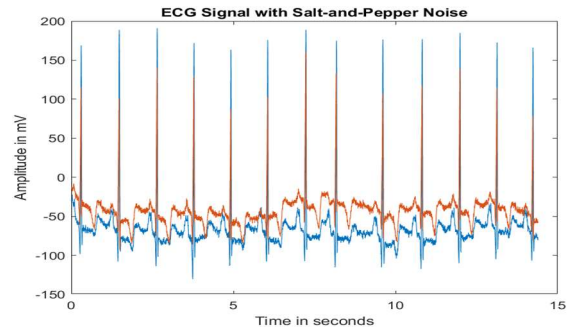


Fig 6.10 Noisy ECG Signal

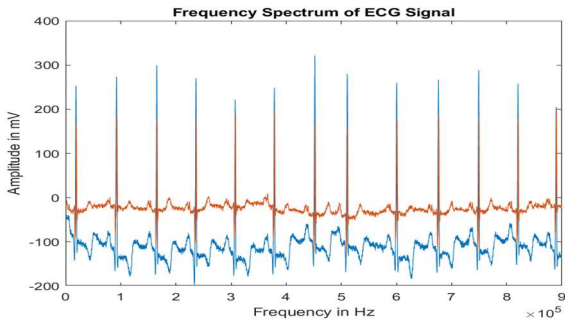


Fig 6.11 : Frequency spectrum

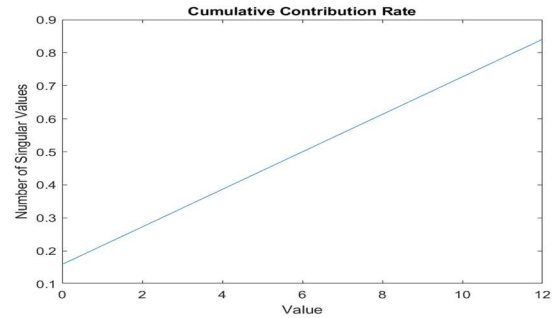


Fig 6.12 : Contribution Rate

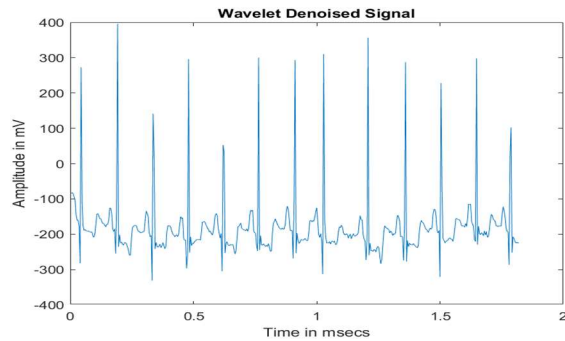


Fig 6.13 : Wavelet Denoised Signal

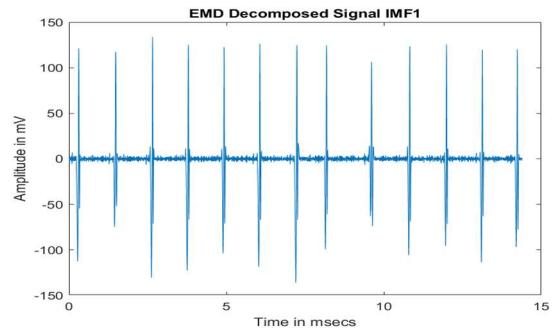


Fig 6.14 EMD Decomposed First IMF

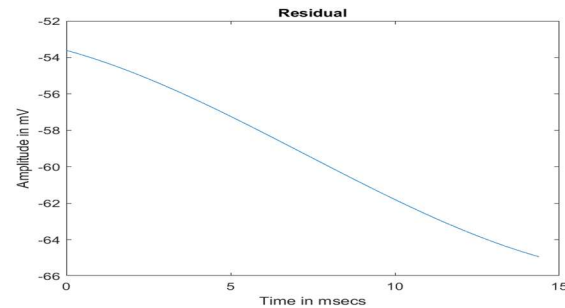


Fig 6.15 Residual Signal

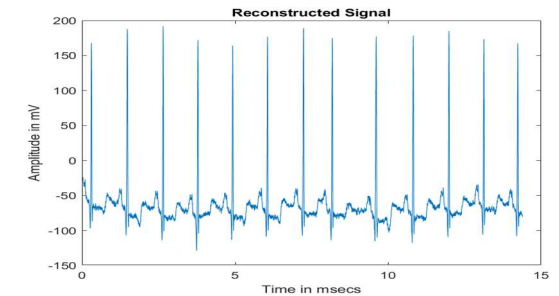


Fig 6.16 :Reconstructed Signal using EMD

Table 6.1 : SNR, MSE, Accuracy measures for Random Noise

Random Noise										
Test sample	Metrics	[1,5]			[1,10]			[1,15]		
		SVD	WAV	EMD	SVD	WAV	EMD	SVD	WAV	EMD
sample 1	SNR	24.32	26.72	27.02	18.50	20.95	21.49	14.92	17.44	18.23
	MSE	0.82	0.0	2.68e-32	0.85	0.010	2.68e-32	0.8794	0.0228	2.68e-32
	ACC	89.32%	89.57%	89.73%	88.91%	89.39%	89.72%	88.44%	89.16%	89.72%
sample 2	SNR	34.44	23.95	24.31	28.91	18.13	18.81	25.70	14.67	15.61
	MSE	8.62	0.003	1.35e-32	8.65	0.010	1.35e-32	8.69	0.02	1.35e-32
	ACC	84.02%	96.79%	96.80%	83.99%	96.78%	96.80%	83.97%	96.72%	96.80%
sample 3	SNR	26.03	27.26	27.47	20.02	21.66	22.02	16.91	18.10	18.61
	MSE	2.14	0.0030	2.67e-32	2.17	0.0106	2.67e-32	2.20	0.023	2.68e-32
	ACC	85.44%	86.53%	86.78%	85.10%	86.29%	86.78%	84.76%	86.04%	86.76%
sample 4	SNR	25.82	26.57	26.81	20.43	21.001	21.42	17.16	17.56	18.15
	MSE	2.83	0.0031	2.49e-32	2.908	0.010	2.49e-32	2.96	0.022	2.49e-32
	ACC	87.33%	89.02%	88.87%	87.18%	89.12%	88.86%	86.97%	89.17%	88.86%
sample 5	SNR	25.07	27.04	27.25	19.68	21.53	21.90	16.21	17.93	18.46
	MSE	2.41	0.0031	3.02e-32	2.47	0.0106	3.02e-32	2.53	0.023	3.02e-32
	ACC	74.15%	80.34%	80.79%	73.60%	79.92%	80.79%	73.06%	79.51%	80.79%
sample 6	SNR	24.61	28.06	28.18	19.14	22.52	22.75	15.93	19.22	19.53
	MSE	2.87	0.0031	2.91e-32	2.90	0.01	2.91e-32	2.97	0.023	2.91e-32
	ACC	87%	86.87%	87.02%	86.75%	86.72%	87.0%2	86.46%	86.55%	87.02%

Table 6.2 : SNR, MSE, Accuracy measures for Salt and Pepper Noise

Salt and pepper Noise										
Test sample	Metrics	0.1			0.5			0.9		
		SVD	WAV	EMD	SVD	WAV	EMD	SVD	WAV	EMD
sample 1	SNR	45.49	47.79	47.79	38.61	40.90	40.90	36.71	39.00	38.98
	MSE	0.79	2.57e-05	2.68e-32	0.79	1.27e-04	2.68e-32	0.79	1.89e-04	2.68e-32
	ACC	89.78%	89.73%	89.73%	89.78%	89.72%	89.72%	89.80%	89.73%	89.72%
sample 2	SNR	54.82	44.67	44.67	48.48	38.33	38.33	46.43	36.31	36.28
	MSE	8.60	2.70e-05	1.35e-32	8.60	1.22e-04	1.35e-32	8.6009	1.93e-04	1.35e-32
	ACC	84.05%	96.80%	96.80%	84.05%	96.80%	96.80%	84.05%	96.80%	96.80%
sample 3	SNR	46.60	47.85	47.85	40.04	41.30	41.30	38.15	39.40	39.40
	MSE	2.12	2.63e-05	2.67e-32	2.11	1.25e-04	2.67e-32	2.11	1.92e-04	2.67e-32
	ACC	85.82%	86.78%	86.82%	85.82%	86.79%	86.79%	85.84%	86.80%	86.79%
sample 4	SNR	46.29	47.25	47.25	39.89	40.86	40.86	37.90	38.86	38.87
	MSE	2.76	2.90e-05	2.49e-32	2.76	1.23e-04	2.49e-32	2.76	1.91e-04	2.49e-32
	ACC	87.49%	88.86%	88.86%	87.49%	88.86%	88.86%	87.50%	88.85%	88.86%
sample 5	SNR	45.42	47.42	47.42	39.26	41.36	41.36	37.14	39.23	39.24
	MSE	2.35	2.85e-05	3.02e-32	2.35	1.23e-04	3.02e-32	2.35	1.94e-04	3.02e-32
	ACC	74.78%	80.79%	80.79%	74.80%	80.80%	80.80%	74.82%	80.82%	80.83%
sample 6	SNR	45.27	48.80	48.80	38.67	42.19	42.19	36.69	40.22	40.22
	MSE	2.80	2.63e-05	2.91e-32	2.82	1.20e-04	2.91e-32	2.81	1.93e-04	2.91e-32
	ACC	87.33%	87.02%	87.02%	87.34%	87.03%	87.34%	87.36%	88.02%	88.02%

Table 6.3 : SNR, MSE, Accuracy measures for Uniform Noise

Uniform noise										
Test sample	Metrics	0.1			0.5			0.9		
		SVD	WAV	EMD	SVD	WAV	EMD	SVD	WAV	EMD
sample 1	SNR	25.47	27.75	27.75	12.07	14.31	14.05	7.48	9.51	8.78
	MSE	0.79	0.0028	2.68e-32	0.85	0.07	2.68e-32	1.04	0.23	2.68e-32
	ACC	89.78%	89.76%	89.72%	88.79%	88.53%	89.72%	87.39%	87.34%	89.72%
sample 2	SNR	21.86	11.95	11.70	8.57	2.09	2.24	4.89	0.94	7.15
	MSE	8.62	0.056	1.35e-32	10.01	1.39	1.35e-31	13.28	4.51	1.35e-32
	ACC	84.01%	96.59%	96.80%	82.79%	92.73%	96.80%	80.38%	87.91%	96.80%
sample 3	SNR	27.83	29.07	29.08	13.98	15.21	15.06	9.55	10.63	10.13
	MSE	2.12	0.0019	2.67e-32	2.16	0.04	2.67e-32	2.35	0.152	2.67e-32
	ACC	85.80%	86.74%	86.78%	85.56%	86.46%	86.79%	85.01%	85.85%	86.78%
sample 4	SNR	22.69	23.64	23.63	9.15	9.90	9.56	5.13	5.61	4.44
	MSE	2.77	0.005	2.49e-32	2.92	0.13	2.49e-32	3.300	0.43	2.49e-32
	ACC	87.45%	88.83%	88.86%	86.75%	88.13%	88.86%	85.28%	86.48%	88.86%
sample 5	SNR	19.67	21.76	21.70	6.45	8.17	7.57	3.33	4.56	2.63
	MSE	2.36	0.01	3.02e-32	2.60	0.23	3.02e-32	3.26	0.85	3.02e-32
	ACC	74.74%	80.7%	80.80%	73.58%	79.45%	80.79%	70.56%	75.0%	80.79%
sample 6	SNR	21.25	24.77	24.75	7.75	11.09	10.56	4.22	6.62	5.45
	MSE	2.82	0.008	2.92e-32	2.94	0.19	2.91e-32	3.42	0.62	2.91e-32
	ACC	87.29%	86.98%	87.02%	86.52%	86.15%	87.02%	85.23%	85.17%	87.02%
sample 7	SNR	28.69	30.22	30.22	14.60	16.08	15.99	9.17	10.53	11.19
	MSE	3.39	0.011	1.60e-31	3.68	0.30	1.60e-31	4.36	0.94	1.60e-31
	ACC	85.96%	86.93%	86.96%	86.87%	86.02%	86.94%	84.83%	84.09%	86.94%

Table 6.4 : SNR, MSE, Accuracy measures for Power Line Interference Noise

Power line interference Noise										
Test sample	Metrics	10			30			50		
		SVD	WAV	EMD	SVD	WAV	EMD	SVD	WAV	EMD
sample 1	SNR	24.14	26.43	26.42	24.14	26.42	26.41	24.14	26.42	26.42
	MSE	0.796	0.0035	2.68e-32	0.80	0.0035	2.68e-32	0.79	0.0035	2.68e-32
	ACC	89.73%	89.67%	89.73%	89.73%	89.68%	89.73%	89.74%	89.68%	89.73%
sample 2	SNR	33.88	23.75	23.73	33.88	23.75	23.73	33.88	23.75	23.73
	MSE	8.60	0.0035	1.35e-32	8.60	0.0035	1.35e-32	8.60	0.0035	1.35e-32
	ACC	84.04%	96.78%	96.80%	84.04%	96.79%	96.80%	84.04%	96.78%	96.80%
sample 3	SNR	25.62	26.85	26.86	25.62	26.87	26.86	25.61	26.87	26.86
	MSE	2.13	0.0035	2.67e-32	2.13	0.0035	2.67e-32	2.11	0.0035	2.67e-32
	ACC	85.82%	86.80%	86.80%	85.80%	86.77%	86.79%	85.80%	86.77%	86.78%
sample 4	SNR	25.33	26.29	26.29	25.33	26.29	26.29	25.34	26.29	26.28
	MSE	2.76	0.0035	2.49e-32	2.77	0.0035	2.49e-32	2.77	0.0035	2.49e-32
	ACC	87.48%	88.85%	88.87%	87.47%	88.85%	88.87%	87.47%	88.84%	88.87%
sample 5	SNR	24.65	26.74	26.73	24.65	26.74	26.73	24.65	26.74	26.73
	MSE	2.35	0.0035	3.02e-32	2.36	0.0035	3.02e-32	2.36	0.0035	3.02e-32
	ACC	74.76%	80.76%	80.80%	74.77%	80.77%	80.79%	74.76%	80.77%	80.80%
sample 6	SNR	24.18	27.68	27.68	24.18	27.69	27.68	24.18	27.69	27.69
	MSE	2.82	0.0035	2.91e-32	2.82	0.0035	2.91e-32	2.82	0.0035	2.91e-32
	ACC	87.33%	87.02%	87.02%	87.32%	87.01%	87.03%	87.32%	87.01%	87.03%
sample 7	SNR	34.06	35.60	35.60	34.07	35.59	35.60	34.06	35.59	35.60
	MSE	3.37	0.0035	1.60e-31	3.39	0.0035	1.60e-31	3.39	0.0035	1.60e-31
	ACC	87.97%	86.83%	86.95%	87.97%	86.93%	89.94%	87.98%	86.94%	86.95%

Table 6.5 : SNR, MSE, Accuracy measures for Baseline Wander Noise

Baseline Wander Noise										
Test sample	Metrics	10			30			50		
		SVD	WAV	EMD	SVD	WAV	EMD	SVD	WAV	EMD
sample 1	SNR	24.14	26.43	26.42	24.14	26.42	26.42	24.14	26.42	26.41
	MSE	0.79	0.0035	2.68e-32	0.801	0.0035	2.68e-32	0.7974	0.0035	2.68e-32
	ACC	89.74%	89.68%	89.73%	89.74%	89.68%	89.73%	89.74%	89.68%	89.73%
sample 2	SNR	33.88	23.75	23.74	33.88	23.74	23.73	33.88	23.73	23.75
	MSE	8.60	0.0035	1.35e-32	8.60	0.0035	1.35e-32	8.60	0.0035	1.35e-32
	ACC	84.04%	96.78%	96.80%	84.04%	96.79%	96.80%	84.04%	96.78%	96.80%
sample 3	SNR	25.62	26.85	26.86	25.62	26.87	26.86	25.61	26.87	26.86
	MSE	2.13	0.0035	2.67e-32	2.13	0.0035	2.67e-32	2.11	0.0035	2.67e-32
	ACC	85.82%	86.8%	86.79%	85.80%	86.77%	86.78%	85.80%	86.77%	86.89%
sample 4	SNR	25.33	26.29	26.29	25.33	26.29	26.29	25.34	26.29	26.28
	MSE	2.76	0.0035	2.49e-32	2.77	0.0035	2.49e-32	2.77	0.0035	2.49e-32
	ACC	87.48%	88.85%	88.87%	87.47%	88.85%	88.87%	87.47%	88.84%	88.86%
sample 5	SNR	24.65	26.74	26.73	24.65	26.74	26.74	24.65	26.74	26.74
	MSE	2.35	0.0035	3.02e-32	2.36	0.0035	3.02e-32	2.36	0.0035	3.02e-32
	ACC	74.76%	80.76%	80.80%	74.76%	80.77%	80.79%	74.76%	80.77%	80.79%
sample 6	SNR	24.18	27.68	27.68	24.18	27.69	27.68	24.18	27.69	27.69
	MSE	2.82	0.0035	2.91e-32	2.82	0.0035	2.91e-32	2.82	0.0035	2.91e-32
	ACC	87.33%	87.03%	87.03%	87.32%	87.01%	87.03%	87.32%	87.01%	87.03%
Sample 7	SNR	34.06	35.59	35.60	34.06	35.59	35.59	34.06	35.59	35.60
	MSE	3.37	0.0035	1.60e-31	3.39	0.0035	1.60e-31	3.39	0.0035	1.60e-31
	ACC	87.97%	86.93%	86.95%	87.97%	86.93%	86.95%	87.98%	86.93%	86.95%

CHAPTER-7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

In this study the feature extraction and denoising the ECG signal based on Empirical Mode Decomposition is proposed and compared the results based on the Singular Value Decomposition and Wavelet. Based on the Observation and experiment the signals based on adding different types of noises (Random Noise, Salt and pepper Noise, Uniform Noise and Power Line Interference Noise) at different levels. In this paper we observe the SNR rate is high for salt and pepper noise compared with other noise. For all the noises which are taken to examine yield almost same range of MSE and accuracy but the proposed signal achieves better mean square error than wavelet and SVD and the proposed signal also has better accuracy. The major disadvantage observed in the terms of wavelet is using other wavelet to denoise which leads to loss the information in cases of taking wrong wavelet, the proposed algorithm has advantage of analyzing or denoising the ECG signal with the help of signal that is reason for obtaining very low MSE.

7.2 FUTURES COPE

In this project, we developed and evaluated techniques for ECG feature extraction and denoising, including Empirical Mode Decomposition (EMD), Singular Value Decomposition (SVD), and wavelet-based methods. By analyzing the performance of these techniques in terms of Signal-to-Noise Ratio (SNR), Mean Squared Error (MSE), and accuracy metrics, we demonstrated their effectiveness in enhancing ECG signal quality and extracting relevant features. Moving forward, future endeavors could focus on algorithm optimization, exploring advanced denoising techniques, enhancing feature extraction methods, conducting clinical validation studies, integrating algorithms into real-time systems, adapting techniques to different ECG types, and exploring novel signal processing approaches. These efforts aim to improve the accuracy, efficiency, and applicability of ECG analysis methods, ultimately benefiting patient care and diagnostic processes in clinical settings.

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APPENDIX

MATLAB CODE: For Random Noise

```
clear
close all;
clc

%Importing Input ECG Signal
% load('118e06m.mat');
[file,path] = uigetfile('*.mat','Select Signal');
Name1 = [path,file];
x = load(Name1);
x = struct2cell(x);
x = cell2mat(x);
fs = 250;
t = 0:(1/fs):(length(x)-1)/fs;
figure;
plot(t,x);
xlabel('Time in msec');
ylabel('Amplitude in mV');
title('Input ECG Signal');

%adding noise
n = randi([1 15],1,length(x));%rand(1,length(x));%awgn(x,snr);
xn = x+n;
tn = 0:(1/fs):(length(xn)-1)/fs;
figure;
plot(tn,xn);
xlabel('Time in msec');
ylabel('Amplitude in mV');
title('Noisy ECG Signal');
Ln = length(n).^2;
%Application of FFT
xf = fft(xn);
xfn = randi([1 10],1,length(xf))+xf;
xfn2 = randi([1 10],1,length(xf))+xf;
xfnn = xf(1,100:1000);
xfn2n = xf(1,1001:2000);
ff = 0:fs:(length(xf)-1)*fs;
figure;
plot(ff,xf);
xlabel('Frequency in Hz');
ylabel('Amplitude in mV');
title('Frequency Spectrum of ECG Signal');
```

%Application of SVD

```
[u, s, v] = svd(xn);  
s_reconstructed = u*s*v';  
s2 = svd(xf);  
figure;  
tsn = 0:(1/fs):(length(s2)-1)/fs;  
tsn = tsn.*(3*10^3);  
plot(tsn, flip(s2));  
xlabel('Value');  
ylabel('Number of Singular Values');  
title('Cumulative Contribution Rate');
```

%Application of Wavelets

```
wname = 'sym4';  
xw = xn(1,:);  
[cA, cD] = dwt(xw, wname);  
[cA1, cD1] = dwt(cA, wname);  
[cA2, cD2] = dwt(cA1, wname);  
final = idwt(cA, cD, 'sym4');  
tscA = 0:(1/fs):(length(cA2)-1)/fs;  
figure;  
plot(tscA, cA2);  
xlabel('Time in msec');  
ylabel('Amplitude in mV');  
title('Wavelet Denoised Signal');
```

%EMD Decomposition

```
[imf, residual] = emd(x(1,:));  
tEmd = 0:(1/fs):(length(imf(:,1))-1)/fs;  
figure;  
plot(tEmd, imf(:,1));  
xlabel('Time in msec');  
ylabel('Amplitude in mV');  
title('EMD Decomposed Signal IMF1');
```

```
tRes = 0:(1/fs):(length(residual)-1)/fs;  
figure;  
plot(tRes, residual);  
xlabel('Time in msec');  
ylabel('Amplitude in mV');  
title('Residual');
```

%reconstructed signal

```
recsig = sum(imf)+residual';  
tRec = 0:(1/fs):(length(recsig)-1)/fs;  
figure;  
plot(tRec,recsig);  
xlabel('Time in msec');  
ylabel('Amplitude in mV');  
title('Reconstructed Signal');  
figure;  
plot(ts4,xfn2n);  
xlabel('Time in msec');  
ylabel('Amplitude in mV');  
title('Signal Component of EMD');
```

%SNR Calculation

```
SNRE = 10 * log10(sum(recsig.^2) / sum(n(1:length(recsig)).^2));  
SNRW = 10 * log10(sum(final.^2) / sum(n(1:length(final)).^2));  
s_reconstructed_resized = s_reconstructed(1:length(x(1,:)));  
SNRs1 = 10 * log10(sum(s_reconstructed_resized.^2) /  
sum(n(1:length(s_reconstructed_resized)).^2));
```

%MSE Calculation

```
MSEE = sum((x(1,:)-recsig).^2)/Ln;  
MSEW = sum((x(1,:)-final).^2)/Ln;  
MSES = sum((x(1,:) - s_reconstructed_resized).^2)/Ln;
```

%Correlation of Signals**% Reshape n and recsig to column vectors**

```
x = reshape(x, [], 1);  
recsig = reshape(recsig, [], 1);  
final = reshape(final, [], 1);  
s_reconstructed = reshape(s_reconstructed, [], 1);
```

%Accuracy Calculation

```
rmse=sqrt(mean((x(1,:)-recsig).^2));  
a=100*(1-rmse/(max(x)-min(x)));  
disp(['AccuracyEMD:',num2str(a),'%'])  
rmse=sqrt(mean((x(1,:)-final).^2));  
b=100*(1-rmse/(max(x)-min(x)));  
disp(['AccuracyWAV:',num2str(b),'%'])  
rmse=sqrt(mean((x(1,:)-s_reconstructed).^2));  
c=100*(1-rmse/(max(x)-min(x)));  
disp(['AccuracySVD:',num2str(c),'%'])
```

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